

The Rhizobial effector NopT targets Nod factor receptors to regulate symbiosis in *Lotus japonicus*

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Abstract

It is well-studied that type-III effectors are essential required by gram-negative bacteria to directly target different cellular pathways from hosts, thus enhancing their infection. However, in the unique context of legume-rhizobium symbiosis, the role of rhizobial effectors in regulating plant symbiotic pathways remains a subject of hypothesized understanding. Here, we show that NopT, a YopT-type cysteine protease from *Sinorhizobium fredii* NGR234 that could directly target symbiotic signaling pathway by associating with two Nod Factor Receptors (NFR1 and NFR5) from *Lotus japonicus*. NopT is the only effector out of 15 effectors from *S. fredii* NGR234 that inhibits cell death triggered by co-overexpression of NFR1 and NFR5 in *Nicotiana benthamiana*. NopT interacts with both NFR1 and NFR5. NopT proteolytically cleaves NFR5 in both *in vitro* and *in vivo* assays, but is later inactivated *via* phosphorylation by NFR1. NopT plays an essential role in mediating rhizobial infection in *L. japonicus*. The truncated NopT, lacking 50 N-terminal amino acids post-autocleavage, retains the ability to cleave NFR5 but no longer interacts with NFR1. Intriguingly, certain *Sinorhizobium* species exclusively harbor this truncated NopT in their genomes. These findings unveil an intricate regulatory interplay between rhizobia and legumes, wherein rhizobial NopT actively cleaves NFR5, and this cleavage process is subsequently suppressed following NopT phosphorylation by NFR1. This discovery presents a compelling narrative of a nuanced regulatory interplay between rhizobia and legumes, highlighting rhizobial effector as a direct mediator targeting symbiotic pathway from plants.

One-sentence summary

The rhizobial effector NopT forms a mutual interaction with plant's two Nod factor receptors, NFR1 and NFR5, to precisely regulate rhizobial symbiosis in *Lout japonicus*

eLife assessment

This **valuable** study reveals how a rhizobial effector protein cleaves and inhibits a key plant receptor for symbiosis signaling, while the host plant counters by phosphorylating the effector. The molecular evidence for the protein-protein interaction and modification is **solid**, though biological evidence directly linking effector cleavage to rhizobial infection is **incomplete**. With additional functional data, this work could have implications for understanding intricate plant-microbe dynamics during mutualistic interactions.

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Introduction

The legume–rhizobial symbiosis stands as one of the most crucial and intricate mutualistic interactions in nature. Its significance extends beyond the supply of the substantial amount of ammonium provided by rhizobia, influencing plant growth and ecosystem sustainability, to serving as a pivotal model for investigating plant–microbe interactions (Yang et al., 2022 [↗](#)). The orchestration of this symbiosis involves the perception of small molecules, such as nod factors (NFs) and surface polysaccharides (Madsen et al., 2003 [↗](#); Kawaharada et al., 2015 [↗](#)), along with the secretion of proteins, including the type three secretion system (T3SS) effectors, to regulate legume–rhizobial symbiosis (Sugawara et al., 2018 [↗](#); Zhang et al., 2021 [↗](#)).

A receptor heterocomplex, consisting of two Lysin–motif receptor kinases (LYKs), plays a central role in recognizing various polysaccharide molecules. In nonlegumes like *Arabidopsis*, AtLYK5 and AtCERK1 (Chitin elicitor receptor kinase) are required for perception of chitin, triggering plant immunity to combat fungal pathogens (Cao et al., 2014 [↗](#)). Conversely, in legumes, structurally similar receptors like *L. japonicus* NF receptor1 (LjNFR1) and LjNFR5 are vital for recognizing rhizobial lipo–chitooligosaccharide NF, initiating symbiotic signaling transduction (Broghammer et al., 2012 [↗](#); Bozsoki et al., 2017 [↗](#)). The intricate regulation of NFRs is highlighted by the instances of hypersensitive cell death induced by their simultaneous overexpression in *Nicotiana benthamiana* leaves (Madsen et al., 2011 [↗](#)) and defense–like responses in *Medicago truncatula* nodules upon overexpression of MtNFP (Nod factor perception, the ortholog of NFR5) (Moling et al., 2014 [↗](#)), perhaps reflecting potential conservation of the plant LYK receptors in interactions with both pathogens and symbionts. Therefore, the protein abundance of NFRs must be tightly controlled to avoid this resistance response, allowing rhizobial infection to proceed, emphasizing the need for tight control to enable rhizobial infection.

T3SS effectors play a critical role in suppressing pattern–triggered immunity (PTI) during pathogen infections in plants (Jones and Dangl, 2006 [↗](#); Schubert et al., 2020 [↗](#)). To avoid diseases, specific resistance (R) genes are employed to recognize successful avirulence (Avr) effectors which enable pathogens overcome PTI, known as effector–triggered immunity (ETI), a much stronger immune response combined with programmed cell death (Jones and Dangl, 2006 [↗](#)). However, in contrast to the well-studied function of Avr effectors from phytopathogenic bacteria in mitigating immunity by different mechanisms, how these effectors from symbiotic bacteria might contribute to mutualistic interaction is largely unknown.

Some effectors, such as YopT-type cysteine protease family, including *Pseudomonas* AvrPphB, *Yersinia* YopT, and *Rhizobial* NopT, are well conserved across different bacteria. AvrPphB from *Pseudomonas syringae* can directly proteolytically cleaves *Arabidopsis* receptor-like cytoplasmic

kinase (RLCK) family proteins including PBS1, PBS1-like proteins, and BIK1, to suppress PTI (Zhang et al., 2010). While, the cleavage of PBS1 serves as a “decoy” in activating RPS5 (RESISTANCE TO PSEUDOMONAS SYRINGAE5)–mediated ETI (Shao et al., 2003b). *Yersinia* YopT directly cleaves Rho family leading to their inactivation to disrupt actin cytoskeleton of host cells (Shao et al., 2003b; Shao et al., 2003a). While, in *Rhizobium* species, NopT have a dual function in regulating symbiosis. The exogenous expression of NopT from *Sinorhizobium fredii* NGR234 in *S. fredii* USDA257 almost blocks nodulation in soybean Nanfeng 15 but not other soybean cultivars (Khan et al., 2022), implying that NopT might act as an Avr effector. However, NopT positively regulates nodule formation in the interaction between *S. fredii* HH103 and soybean Suinong 14 (Li et al., 2023). NopT from *Bradyrhizobium* sp. ORS3257 promotes rhizobial infection in nodules of *Aeschynomene indica* (Teulet et al., 2019). Despite this, little is understood about how NopT precisely modulates rhizobial infection and the variations among NopTs from different rhizobia strains.

The challenge in studying rhizobial effectors lies in the availability of suitable assays by which to characterize their phenotypes. In this study, we capitalized on the observation that ectopic expression of NFR1 and NFR5 in *N. benthamiana* leaves induces programmed cell death (Madsen et al., 2011), a response that might be modulated by rhizobial effectors. Among the 15 known T3SS effector proteins encoded within the *S. fredii* NGR234 genome, only NopT, a homolog of the pathogenic effectors, YopT and AvrPphB (Shao et al., 2002; Dai et al., 2008), could specifically suppress this cell death response. This suggests that NopT might be an effector associated with NFRs. Our findings demonstrate that NopT interacts with both NFR1 and NFR5 at the plasma membrane, and cleaves NFR5 proteolytically at the juxtamembrane position to suppresses NF-mediated physiological responses. Perhaps as a means to regulate NopT activity, NFR1 phosphorylates intact NopT, inactivating its protease activity. However, the autocleaved NopT retains the ability to cleave NFR5 but loses its association with NFR1. Intriguingly, some *Sinorhizobium* species possess only the truncated versions of NopT (relative to NGR234 NopT) that might evade suppression by NFR1. This study unveils a distinct and mutual regulatory model, in which the rhizobial effector NopT directly targets NFR5 to suppress symbiotic signaling, and NFR1-mediated phosphorylation acts as a negative feedback mechanism. The model provides essential insights into the intricate interplay between legumes and rhizobia, revealing a counteraction where a rhizobal T3SS effector impairs NF signaling by directly targeting NFR5, and NFR1 phosphorylates and inactivates the rhizobal T3SS effector in response.

Results

NopT suppresses the cell death response from overexpression of NFR1 and NFR5

To examine the role of effector proteins in regulating rhizobial infection, *S. fredii* NGR234, a strain with exceptionally broad host-range was chosen. Initial experiments were performed to screen all 15 effector proteins identified from *S. fredii* NGR234 (Kimbrel et al., 2013), for their ability to suppress or enhance the cell death response in *N. benthamiana* leaves due to co-expression of LjNFR1 and LjNFR5 (Fig. S1A–S1C). Out of the effectors tested, only NopT, a cysteine protease belonging to the YopT family (Shao et al., 2002; Dai et al., 2008), could suppress LjNFR1 and LjNFR5-induced cell death (Fig. 1A and S1D–S1F). Cysteine-93 of NopT is a known residue essential for its protease activity (Dai et al., 2008). In contrast to the wild-type NopT protein, expression of NopT^{C93S}, an inactive version, was unable to suppress cell death induced by NFR1 and NFR5 compared with controls (Fig. 1B), indicating the protease activity of NopT is essential for suppression of cell death in *N. benthamiana*. In contrast to the suppressive function on cell death, NopM, an effector protein from *S. fredii* NGR234 reported to have an E3 ligase activity, induced strong cell death when overexpressed in *N. benthamiana* leaves (Fig. 1C).

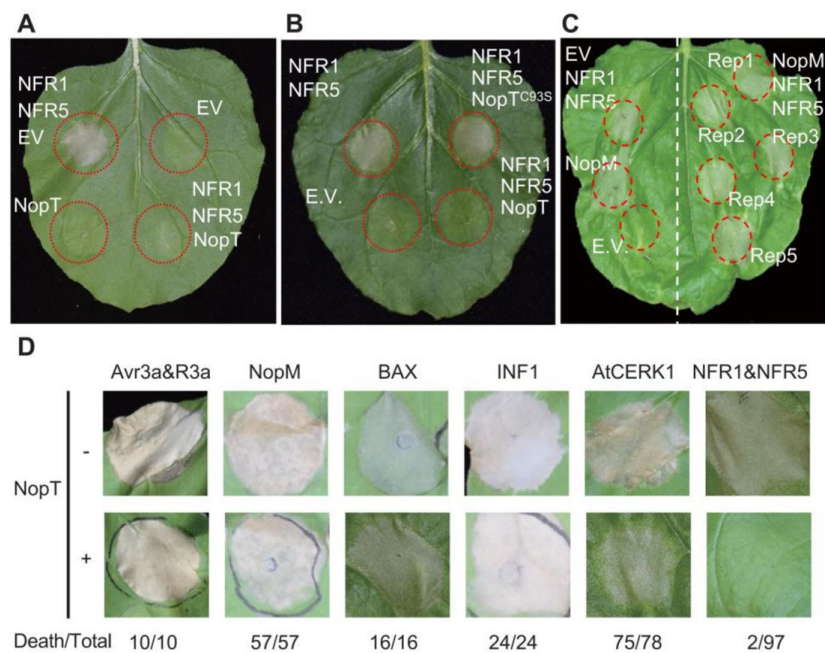


Figure 1.

NopT specifically suppresses cell death triggered by NFR1 & NFR5 in *N. benthamiana*. *Agrobacterium* strains harboring genes were NopT or EV infiltrated into *N. benthamiana*, 12 hpi followed by infiltration with *agrobacterium* containing NFR1 and NFR5. Different leaf discs indicate where different proteins were expressed. NopT (A) but not NopT^{C93S}, the protease-inactive version of NopT (B) suppressed cell death induced by expression of NFR1 and NFR5. (C) Expression of NopM induced cell death in *N. benthamiana*. (D) Expression of NopT could not suppress the cell death triggered by expression of Avr3a&R3a, NopM, BAX, INF1 or AtCERK1. Digital numbers in (D) represent the number of leaf discs with cell death and total leaf discs tested. All pictures in the Figure are representative of at least three independent biological replicates.

As controls, we tested whether the action of NopT was specific for NFR1/NFR5 expression or whether it could act generally to suppress cell death. A variety of other proteins have been reported to induce cell death when expressed in *N. benthamiana* leaves; specifically, Avr3a and R3a (Bos et al., 2006), apoptosis regulator BAX (Bcl-2-associated X protein) (Gan et al., 2009), INF1, an elicitor from *Phytophthora infestans* (Vleeshouwers et al., 2006), Arabidopsis AtCERK1 (Cao et al., 2014), and NopM (Fig. 1C). In contrast to the case with LjNFR1 and LjNFR5, expression of NopT was unable to suppress the cell death induced when each of these proteins was expressed in tobacco leaves (Fig. 1D). Hence, the effect of NopT appears to be specific for the NF receptors. These results suggested a hypothesis that rhizobial NopT protease might act directly on the NFR1 and/or NFR5 proteins.

NopT colocalizes with and interacts with NFR1 and NFR5 at the plasma membrane

NopT contains an acylation site that appears to localize NopT to cellular membranes (Downen et al., 2009). Therefore, we tested the possibility that NopT, NFR1 and NFR5 co-localize on the plasma membrane. *In vivo* bimolecular fluorescence complementation (BiFC) assays were used to examine the colocalization of NopT with LjNFR1 and LjNFR5. NopT, LjNFR1 and LjNFR5 were all C-terminally tagged with nYFP and cYFP. Fluorescence signals representing the associations between NopT and LjNFR1, LjNFR5, as well as itself, were detected at the plasma membrane (Fig. 2A and 2B), suggesting that NopT colocalized with and interacted with both LjNFR1 and LjNFR5.

Interactions between NopT and LjNFR1 and LjNFR5 were further verified using a Split-luciferase (Split-LUC) assay (Fig. 2C). Co-expression of NopT-cLUC and LjNFR1-nLUC or LjNFR5-nLUC produced strong luminescence signals in leaf discs compared with the negative control expressing FLS2-nLUC and NopT-cLUC (Fig. 2C).

It has been known that the acylation of NopT could change its subcellular localization (Downen et al., 2009). We then examined the interaction between NFR1/NFR5 and NopT^{G50A/C51A/C52A}, a mutant version on the acylation sites. The interactions between NopT^{G50A/C51A/C52A} and NFR1/NFR5 were significantly reduced compared to NopT^{WT} in Split-LUC assays (Fig. S2A and S2B). While the mutation at catalytic site of NopT (NopT^{C93S}) remained similar interaction with NFR1 and NFR5 (Fig. S2C). The interaction results were further confirmed by co-immunoprecipitation (co-IP) assay. The C-terminally HA-tagged LjNFR1 and LjNFR5 could be precipitated by anti-FLAG antibody in the sample expressing NopT-FLAG compared with controls (Fig. 2D). The interaction between NopT and LjNFR1 and LjNFR5 suggested that LjNFR5 or LjNFR1 might be a target protein for proteolysis by NopT.

NopT proteolytically cleaves LjNFR5 at the juxtamembrane domain

A previous study showed that NopT protease activity is detectable upon autocleavage of about 50 amino acids from its N-terminus (Dai et al., 2008). We next tested whether NopT could directly proteolyze LjNFR1 and/or LjNFR5 when co-expressed in *N. benthamiana* leaves. The addition of NopT but not NopT^{C93S}, an inactive version, resulted in proteolytic cleavage of LjNFR5-GFP but not LjNFR1-GFP as shown by immunoblotting of the SDS-PAGE gel (Fig. 3A and S3A). The *in vivo* cleavage assay in *L. japonicus* transgenic root also showed that NopT but not NopT^{C93S} resulted in proteolytic cleavage of LjNFR5 (Fig. 3B). Based on the molecular weight change, the cleavage sites were predicted to occur within the cytoplasmic domain (CD) of LjNFR5. Therefore, the CD of NFR5 was constructed as a recombinant protein (Strep-NFR5^{CD}-HA) and then co-expressed with NopT-FLAG or the inactive version NopT^{C93S}-FLAG in *Escherichia coli* cells. The results showed that the active NopT protease could cleave NFR5^{CD} resulting in a band roughly ~5 kDa less than the control as detected using anti-HA antibody (Fig. 3D). In the same experiment, when the CD of NFR1 was co-expressed with NopT, no cleavage product was detected in immunoblotting assay.

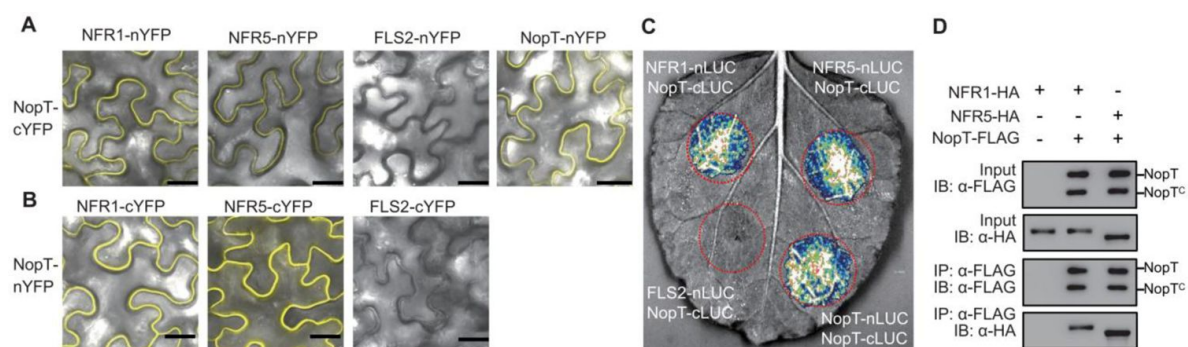


Figure 2.

NopT interacts with NFR1 and NFR5.

Interactions between NopT and NFR1 and NFR5 were detected using Split-YFP assay (A) and (B), Split-LUC (C) and co-IP (D) assays in *N. benthamiana*. (A) and (B). nYFP and cYFP tags were fused at the C-terminus of NopT, NFR1, NFR5, and negative control FLS2. YFP fluorescence signals represent protein-protein interactions. Scar bar=25 μm. (C). nLUC and cLUC tags were fused at the C-terminus of NopT, NFR1, NFR5, and negative control FLS2. Luminescence signals represent protein-protein interactions. (D). HA-tagged NFR1 and NFR5 and FLAG-tagged NopT were expressed in plant cells followed by immunoprecipitation using anti-FLAG antibody and determined by immunoblot using anti-HA and anti-FLAG antibodies. All data in Figure are representative of three biological replicates.

(Fig. 3C). However, migration of the bands representing NopT or NopT^{C93S} was significantly retarded as seen after immunoblotting (Fig. 3C). This retardation of NopT protein band may be the result of phosphorylation by NFR1.

The cleavage product with ~5 kDa lower molecular weight in the cleavage assay, suggested that the cleavage site is located within the juxtamembrane (JM) position of NFR5. To test this hypothesis, SUMO-NFR5^{KD}-HA, a recombinant protein with the JM of NFR5 replaced with a SUMO tag, was constructed and used for the cleavage assay.

Indeed, NopT could not cleave SUMO-NFR5^{KD}-HA (Fig. S3B). To confirm this conclusion, the JM of NFR5 was used to bridge SUMO and GFP tags to create a recombinant protein, SUMO-JM-GFP. *In vitro* cleavage assay, strong band intensity representing GFP was detected in the flow though sample incubated with NopT but not NopT^{C93S} (Fig. 3E). These data strongly indicated that the cleavage site for NopT is located at the JM of NFR5.

The cleavage of LjNFR5 at JM by NopT is dependent on multiple residues of NFR5

NopT has an autocleavage site within its N-terminus, we identified the peptide (DKLLSGV) from Asp-288 to Val-294 of NFR5 (Fig. S4B) as showing the highest similarity with the NopT autocleavage region (DKMGCCA). However, the mutant version of NFR5^{CD} with the peptide (DKLLSGV) replaced with 7 alanines could still be cleaved by NopT (Fig. S3C). NopT and its homolog AvrPphB (Shao et al., 2003b), an effector from the pathogen *Pseudomonas syringa*, belong to the same protein family.

Hence, we set up a control experiment by constructing a version of NFR5^{CD}, where the “DKLLSGV” motif was replaced with the cleavage sites of AvrPphB. In the cleavage assay, we found that this modified NFR5^{CD} was cleaved by both AvrPphB and NopT, and the cleaved product of modified NFR5^{CD} by AvrPphB shifting faster than by NopT (Fig. S3C), suggesting that the cleavage site of NFR5 by NopT located ahead of “DKLLSGV” motif. As mentioned above, NopT undergoes autocleavage but it is not clear whether this is due to inter-proteolysis or intra-proteolysis. His-205 and Aps-220 of NopT are two other conserved residues required for the protease activity of YopT, another homolog of NopT from *Yersinia* whose function has been extensively studied (Shao et al., 2003b). Therefore, we targeted these residues for site-directed mutagenesis. The cleavage assay showed that NopT could not proteolyze NopT^{C93S}, NopT^{H205A}, or NopT^{D220A} (Fig. S3D), suggesting that intra-proteolysis might occur and the autocleavage site of NopT could not be used to predict the cleavage site of substrate.

To define the cleavage site of NFR5, we created a variety of point mutations covering the JM domain of NFR5 (Val-269 to Cys-320) and first seven a.a. of the kinase domain (Lys-321 to Tyr-327). First, 17 mutant versions of NFR5^{JM} were created with three adjacent amino acids replaced with three alanines for proteolysis assay in *E. coli* cells. However, all 17 mutant versions of NFR5^{CD} could be cleaved by NopT but not NopT^{C93S} (Fig. S3E-S3H). An additional 7 truncated versions of NFR5^{JM}, with deletion of 10 adjacent residues, were created and tested for cleavage by NopT. Surprisingly, the cleavage products were still observed from all 7 truncated versions of NFR5^{JM} (Fig. S3I). The *in vitro* cleavage assay suggested that the JM domain might have multiple sites susceptible to be cleaved by NopT, which is different from the specific cleavage site identified for its homologous protein AvrPphB. In a parallel experiment, the recombinant protein, His-SUMO-NFR5^{CD}-GST expressed in *E. coli* was subjected for N-terminal protein sequencing determined by Liquid Chromatography Mass Spectrometry analysis after cleavage. Based on alignment using identified peptides, the cleavage site of NFR5 by NopT was mapped to be after four basic amino acids in the JM of NFR5, “RRKK” from 271 to 275 a.a. of NFR5 (Fig. S4C). However, the result is not consistent with the data from cleavage assay. When the region from Tyr-268 to Leu-277 was

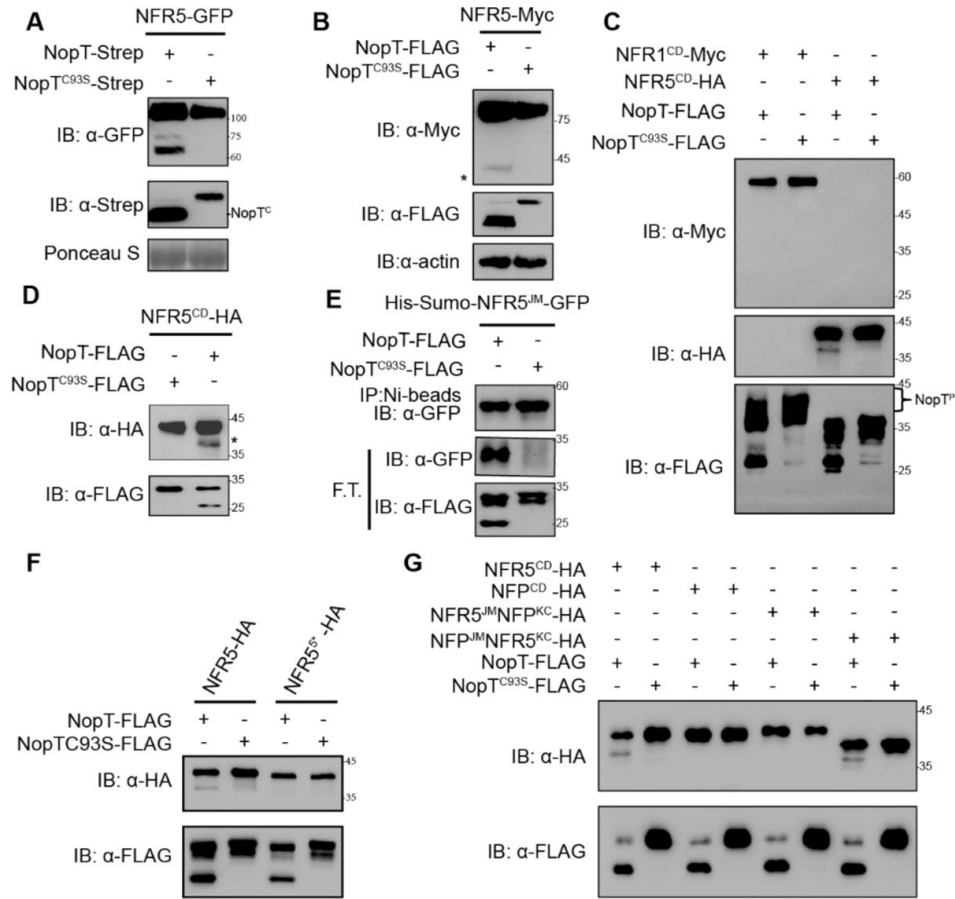


Figure 3.

NopT proteolyzes NFR5 at its juxtamembrane domain.

(A) NopT but not NopT^{C93S} cleaves NFR5-GFP protein expressed in *N. benthamiana* cells by releasing NFR5^{CD}-GFP. (B) NopT but not NopT^{C93S} cleaves NFR5-Myc protein expressed in *L. japonicus* transgenic root by releasing NFR5^{CD}-Myc. Asterisk indicated the cleaved variant of NFR5. (C) A repeated experiment testing the cleavage of NFR5^{CD} but not NFR1^{CD} fused with different tags. Asterisk represents the phosphorylated NopT by inducing a band retardation on SDS-PAGE gel. (D) NopT but not NopT^{C93S} cleaves the CD of NFR5 protein expressed in *E. coli* cells by releasing the kinase domain of NFR5. Asterisk indicated the cleaved variant of NFR5. (E) NopT cleaves a recombinant protein where His-tagged SUMO and GFP is bridged with the JM domain of NFR5. F.T. represents flow through sample after Ni-beads purification. (F) NopT could not cleave a mutant version of NFR5^{CD}-HA with 5 residues in the JM substituted into other residues (S283Y, G294Q, Y303S, A310I, T311Y). (G) NopT could not cleave *Medicago truncatula* NFP and from NFR5^{JM}-NFP^{KD}. CD represents cytoplasmic domain, KC represents kinase domain and C-terminal tail region, NopT^C indicates the truncated version of NopT after autocleavage by releasing about 50 a.a. at its N-terminus.

deleted, NopT could still proteolyze NFR5-CD (**Fig. S4I**). The cleavage assay suggested that NopT might prefer to proteolyze NFR5 after four polybasic residues in the NFR5^{JM}, but other sites could also be cleaved by NopT.

Because LjNFR5 belongs to a subfamily of plant LYK family proteins that lacks kinase activity (Yang et al., 2022), we then asked whether NopT could proteolyze other LYK proteins in this subfamily, which might be helpful to characterize the biochemical function of NopT. Hence, the CDs from Arabidopsis AtLYK5, *L. japonicus* LjLYS11 and *M. truncatula* MtNFP were used for cleavage assays in the presence of NopT. As shown in **Fig. S3J**, NopT, but not NopT^{C93S}, could cleave the CDs of both AtLYK5 and LjLYS11. Replacement of NFR5^{JM} with the JM domains from either AtLYK5 or LjLYS11 also resulted in cleavage by NopT (**Fig. S3J**). The data strongly indicated that NopT could cleave both AtLYK5 and LjLYS11 at their JM domains. Based on sequence analysis of the JM region of NFR5 and its homologous proteins, five conserved residues (Ser-283, Gly-294, Tyr-303, Ala-310, Thr-311) were identified and mutated for the proteolysis assay (**Fig. S4B**). As shown in **Fig. 3F**, the mutant version of NFR5 (i.e., NFR5^{CD} with five amino acid mutations) could not be cleaved by NopT. Interestingly, NopT could not proteolyze MtNFP^{CD} (**Fig. 3G**). Then, we generated two recombinant proteins, by switching the JM between NFR5^{CD} and NFP^{CD}, generating NFR5^{JM}-NFP^{KC} (KC represents kinase domain and C-terminal tail region) and NFP^{JM}-NFR5^{KC} for *in vitro* cleavage assays. As shown in **Fig. 3G**, NopT could proteolyze NFP^{JM}-NFR5^{KC} but not NFR5^{JM}-NFP^{KC}. The data suggested that the cleavage of NFR5 by NopT might be dependent on the conformational structure of NFR5^{KC} and/or other specific amino acids in NFR5^{KC}. To define this, two additional recombinant proteins with large switch between NFR5^{CD} and NFP^{CD}, NFR5²⁶⁸⁻⁴⁴⁵-NFP⁴⁵⁸⁻⁵⁹⁵ and NFP²⁷⁰⁻⁴⁵⁷-NFR5⁴⁵⁶⁻⁵⁹⁵, were constructed for proteolysis (**Fig. S3K**). However, both of them could be cleaved by NopT, suggesting that a much large region on NFR5^{KC} is required for being proteolyzed by NopT. Based on all these cleavage results, we concluded that NopT could proteolyze NFR5 at the JM domain, which nonadjacent multiple residues within the NFR5^{JM} and the specific KC region are required for the proteolysis.

NopT phosphorylation by LjNFR1 suppresses its proteolytic activity

The perception of rhizobia NFs by the heterocomplex of NFR1 and NFR5 involves the trans-phosphorylation from the NFR1 kinase to the pseudo-kinase NFR5 (Madsen et al., 2011). Rhizobial effector NopT associates with both NFR1 and NFR5, which suggests that NopT might be a phosphorylation target of NFR1 and involved in the transphosphorylation process between the two receptors. Kinase assays showed that NFR1-CD but not the kinase-inactive NFR1^{K351E}-CD (Lys-351 is the conserved residue required for ATP binding) could phosphorylate NopT (**Fig. 3C** and **4B**) and NFR5-CD (**Fig. S5**) as shown by band retardation on SDS-PAGE gels identified using immunoblotting. The band representing phosphorylated NopT shifted upon the addition of a commercial phosphatase, calf intestinal alkaline phosphatase (**Fig. 4B**), indicating that NopT as well as NFR5^{CD} are phosphorylation targets of the NFR1 kinase. NopT autocleavage also gives two bands and the site mutant of NopT at C93S did not affect the interactions between NFRs (**Fig. S2C**). To avoid multi-bands interference of cleaved NopT, NopT^{C93S} was expressed in both wild type plants and *nfr1* mutant then treated with rhizobia or water. After analyzed by Zn²⁺-Phos-tag PAGE gel, several shifted bands were strengthened after rhizobia treatment in WT roots but not in the *nfr1* mutant roots (**Fig. 4A**), suggesting that the phosphorylation of NopT could be induced by NFR1 and multi-sites of NopT were phosphorylated *in planta*. The phosphorylation sites of NopT were further identified using Liquid Chromatography Mass Spectrometry (**Table S1**). We then addressed whether NopT phosphorylation by NFR1 regulates its proteolytic activity on NFR5. All the potential phosphorylation sites of NopT were substituted into either alanine in order to block phosphorylation of these sites or aspartate, which mimicks the acidic condition representing a phosphorylated status. These mutant proteins were then used in the cleavage assay. Aspartate substitutions on three phosphorylation sites of NopT, Thr-97, Ser-119, and Thr-257, resulted in the loss of the ability of NopT to cleave NFR5-CD, as well as the ability to autocleave (**Fig. 4C**). In

contrast, NopT proteins with alanine substitutions on these three sites retained the ability to cleave NFR5-CD and to autocleave (**Fig. 4C**). The data suggest that phosphorylation of NopT by NFR1 likely inhibits proteolytic activity, retaining a fully functional NFR5 and a full length NopT.

NopT dampens rhizobial infection

The fact that *S. fredii* NGR234 NopT could directly target and cleave NFR5 indicates that NopT may be an important player regulating rhizobial infection in legumes. Nod factor signal is known to function in the early stage of rhizobial infection and nodulation specificity (Geurts et al., 2005; Wang et al., 2012). Prior work showed that type three secretion system of *S. fredii* NGR234 might not regulate nodule numbers few weeks after inoculation in *L. japonicus* (Ausmees et al., 2004). Therefore, we focused on the early stage of rhizobial infection. Indeed, inoculation with GFP-labelled *S. fredii* NGR234 Δ nopT induced a massive infection within the root hairs of *L. japonicus* (Gifu) plants compared with the favorable infection by the NGR234 wild type strain (**Fig. 5A** and **5C**). Transgenic expression of NopT in *L. japonicus* root hair reduced the infection of *Mesorhizobium loti* MAFF303099 (**Fig. 5G**). In *L. japonicus* plants transgenically expressing *pNIN::GUS*, greater range of GUS expression (**Fig. 5B** and **5D**) and formation of more nodule primordia (**Fig. 5E**) also reflected a stronger plant response to infection by NGR234 Δ nopT relative to the wild-type strain. To confirm the massive infection is due to the loss of NopT, NopT was expressed in the *S. fredii* NGR234 Δ nopT strain and used for inoculation in *L. japonicus*. Complementation with NopT in NGR234 Δ nopT reduced the massive infection foci to the similar levels as the WT strain (**Fig. 5F**). We then tested whether the proteolytic activity of NopT is required for rhizobial infection.

The protease-dead version of NopT^{C93S} and three phosphorylation mimic versions of NopT, i.e., NopT^{T97D}, NopT^{S119D}, and NopT^{T257D}, which lack proteolytic activity, were individually expressed in NGR234 Δ nopT. Compared with the high levels of infection foci induced by NGR234 Δ nopT in *L. japonicus* plants, inoculation with these four mutant strains induced much lower levels of infection, albeit slightly higher than that induced by the wild-type strain (**Fig. 5F**). Inoculation with NGR234 Δ nopT produced more nodule primordia compared with wild-type strain inoculation 14 dpi (**Fig. 5E**). To verify whether the cleavage of NFR5 reduce rhizobial infection, NFR5-WT and NFR5^{5m}, a mutant version with 5 amino acids substitutions at JM that is resistant to NopT cleavage, were expressed in the *nfr5-3* roots under the control of its native promoter. However, expression of NFR5^{5m} was not observed to rescue the infection phenotype of *nfr5-3* mutant plants (**Fig. S5A**). The possible reason is that amino acids substitutions at JM might lead to a conformation change of NFR5 that is required for its function in symbiotic signaling transduction. Therefore, NFR5-NFP^{KC}, the KC domain of NFR5 replaced with KC domain of NFP, was expressed in the roots under the control of native promoter (**Fig. S5B**). In the *nfr5-3* roots expressing NFR5-NFP^{KC}, no difference of rhizobial infection was observed between *S. fredii* NGR234 Δ nopT and *S. fredii* NGR234 wild type strain, suggesting that the cleavage of NFR5 reduced the rhizobial infection levels. All these data strongly indicated that the protease activity of NopT, the phosphorylation status of NopT, and the proteolysis of NFR5 are important for rhizobial infection.

The truncated version NopT ^{Δ N50} cleaves NFR5 but fails to interact with NFR1

The significance of rhizobial NopT in mediating rhizobial infection led us to analyze its homologs from other rhizobium species. Sequence comparisons based on the amino acids of NopT homologs showed that *Sinorhizobium* and *Bradyrhizobium* were located in different clades (**Fig. S7**). *B. diazoefficiens* (USDA110) is a typical *Bradyrhizobium* strain used to study the symbiotic interaction with soybean. Two NopT homologous genes, named *NopT1*_{USDA110} and *NopT2*_{USDA110} were identified from the USDA110 genome (Fotiadis et al., 2012). However, when tested, neither NopT1_{USDA110} or NopT2_{USDA110} could not cleave NFR5 compared to NopT_{NGR234} (**Fig. 6A**). The possible scenario is that NopT homologs from *Bradyrhizobium* species might have lost the ability to

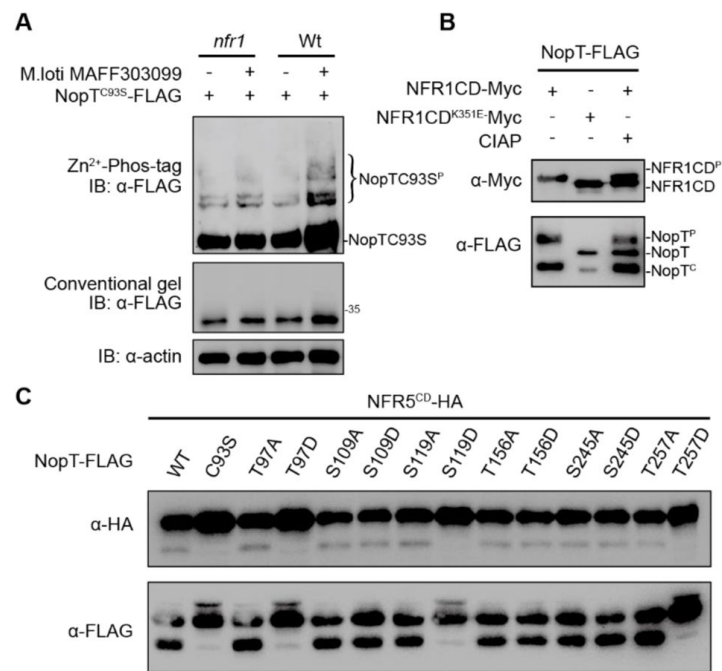


Figure 4.

Phosphorylation of NopT by NFR1 suppresses its proteolytic activity.

(A) *In vivo* phosphorylation assay in *L. japonicus* roots. Phosphorylation of NopT^{C93S} was induced with rhizobia treatment and dependent on *nfr1*. (B) NFR1^{CD} phosphorylates NopT by inducing a gel band shift determined by immunoblotting. (C) The phosphorylation sites on NopT identified by liquid chromatography–mass spectrometry (LC–MS) were either substituted into alanine (A) or aspartate (D) and subsequently tested for their ability to proteolytically cleave NFR5^{CD}.

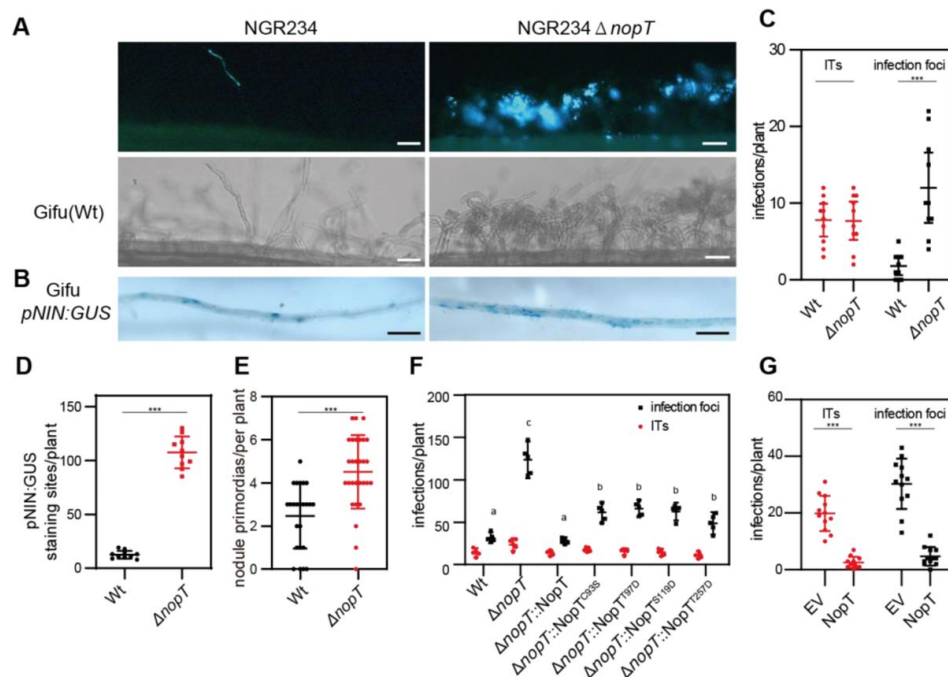


Figure 5.

NopT regulates rhizobial infection in *L. japonicus*.

(A) Rhizobial infection in *L. japonicus* using GFP-labelled WT and *nopT* mutant strains of *S. fredii* NGR234. Fluorescent, GFP-expressing bacteria are imaged in Cyan. Scar bars=12.5 μ m. (B) GUS staining pictures from the roots of *L. japonicus* transgenically expressing GUS under *pNIN* promoter inoculated with WT and *nopT* mutant strains of NGR234. Scar bars=1mm (C) Statistical analysis of rhizobial infection in A (n=10, Student's *t*-test: $P < 0.01$). (D) Statistical analysis of GUS staining sites in B (n=10, Student's *t*-test: $P < 0.01$). (E) Nodule primordia number on the roots 14dpi after inoculation WT or *nopT* mutant strains (n>20, Student's *t*-test: $P < 0.01$). (F) Infection foci number from the wild type plant roots inoculated with WT strains and *nopT* mutant strains expressing NopT with different variations (n=5, Student's *t*-test: $P < 0.01$). (G) Statistical analysis of rhizobial infection in *L. japonicus* transgenic root expressing GFP (EV, empty vector control) or NopT using DsRed-labeled *M. loti* MAFF303099 (n=8, Student's *t*-test: $P < 0.01$).

cleave NFR5 and/or may act on other, unknown cellular target proteins. The genome of some *Sinorhizobium* species, for example *S. fredii* USDA257 and *S. fredii* HH103, encode NopT homologous genes predicted to produce same truncated proteins with a 57 aa deletion from its N-terminus, relative to the NGR234 NopT sequence, which is almost the same as the truncated NopT_{NGR234} after autocleavage. When these NopT homologs (i.e., NopT_{USDA257} and NopT_{HH103}) were tested, both were observed to have the ability to cleave NFR5 (**Fig. 6A**). However, these NopT homologs could not suppress the cell death triggered by NFR1 and NFR5 in *N. benthamiana* leaves (**Fig. 6B**) and failed to complement the massive infection of NGR234ΔnopT (**Fig. 6C** and **6D**), suggesting that NopT_{USDA257} and NopT_{HH103} retained the same catalytic activity as NopT_{NGR234} but might have different substrates. We then asked whether the truncated version of NopT^{ΔN50}, which mimics the truncated NopT after autocleavage, retained the ability to interact with NFR1. However, after repeated tests, the direct interaction between NopT^{ΔN50} and NFR5 could be detected but not the corresponding interaction with NFR1 using Split-LUC assays (**Fig. S2D**). The data suggest that some *Sinorhizobium* species, for example, *S. fredii* NGR234, *S. fredii* USDA257 and *S. fredii* HH103, use a truncated NopT to regulate the rhizobial infection process in a way that avoids negative regulation by NFR1 phosphorylation, which is not true of all rhizobial strains that possess NopT homologs.

Discussion

Plant-microbe interactions involve two sophisticated and intelligent organisms, each employing distinct “weapons” or strategies to communicate mutually, thereby securing more nutrients for their survival (Jones and Dangl, 2006; Schubert et al., 2020). A well-established regulatory model arises from the study of plant-pathogen interactions, where T3SS effectors from gram-negative bacteria are injected into plant cells to directly target and suppress the plant’s immune pathway, favoring pathogenic bacterium infection (Jones and Dangl, 2006). However, the unique model of mutual regulation between legumes and rhizobia in symbiotic interactions remains largely unknown. The broad goal of this study was to offer essential insights into the mutual regulation mechanisms whereby rhizobial effectors directly target symbiotic signaling pathways to dampen their infection in host cells.

Understanding how plants recognize different microbes as “friends” or “foes” is a central question in biology. As the first line of plant response to microbes, the perception of microbial signals by receptor proteins from plants is essential for inducing different signaling pathways to establish various interaction patterns—either symbiotic, pathogenic, or no responses (Zipfel and Oldroyd, 2017). Therefore, it is not surprising to discover that the direct targeting of these receptors by bacterial effectors represents an efficient strategy to manipulate plant signaling pathways for enhancing bacterium infection. In the well-studied *Arabidopsis*–*Pseudomonas syringa* interaction model, the flagellin receptor FLS2 (Flagellin Sensing 2) and its coreceptor BAK1 (BR1-[Brassinosteroid insensitive 1]-associated receptor kinase 1), triggering plant immunity against pathogenic infection, are targeted by several effectors from bacteria (Zipfel et al., 2004; Sun et al., 2013). For example, AvrPto and HopB1 could suppress the kinase activity and cleave the *Arabidopsis* receptor complex, respectively (Xiang et al., 2008; Li et al., 2016). AvrPtoB, an E3 ligase in *P. syringae*, targets FLS2 to promote its degradation, resulting in decreased immune responses in plant cells (Goehre et al., 2008). Considering the evolutionary importance of effectors in infection, it is rational to identify that the rhizobial effector NopT, a member of the C58 protease family, can directly target NFRs, manipulating symbiotic signaling pathways in plants. This was demonstrated using a unique screening system in *N. benthamiana* leaves (**Fig. S1A–C**), in which co-expression of two symbiotic receptors, NFR1 and NFR5, could lead to plant cell death to identify NopT, the only effector from *S. fredii* NGR234. *S. fredii* NGR234ΔnopT mutant strains hyperinfect *L. japonicus* roots, and overexpression of NopT in the roots reduces rhizobial infection (**Fig. 5A–5D**), consistent with the observed suppression of cell death in *N. benthamiana* leaves

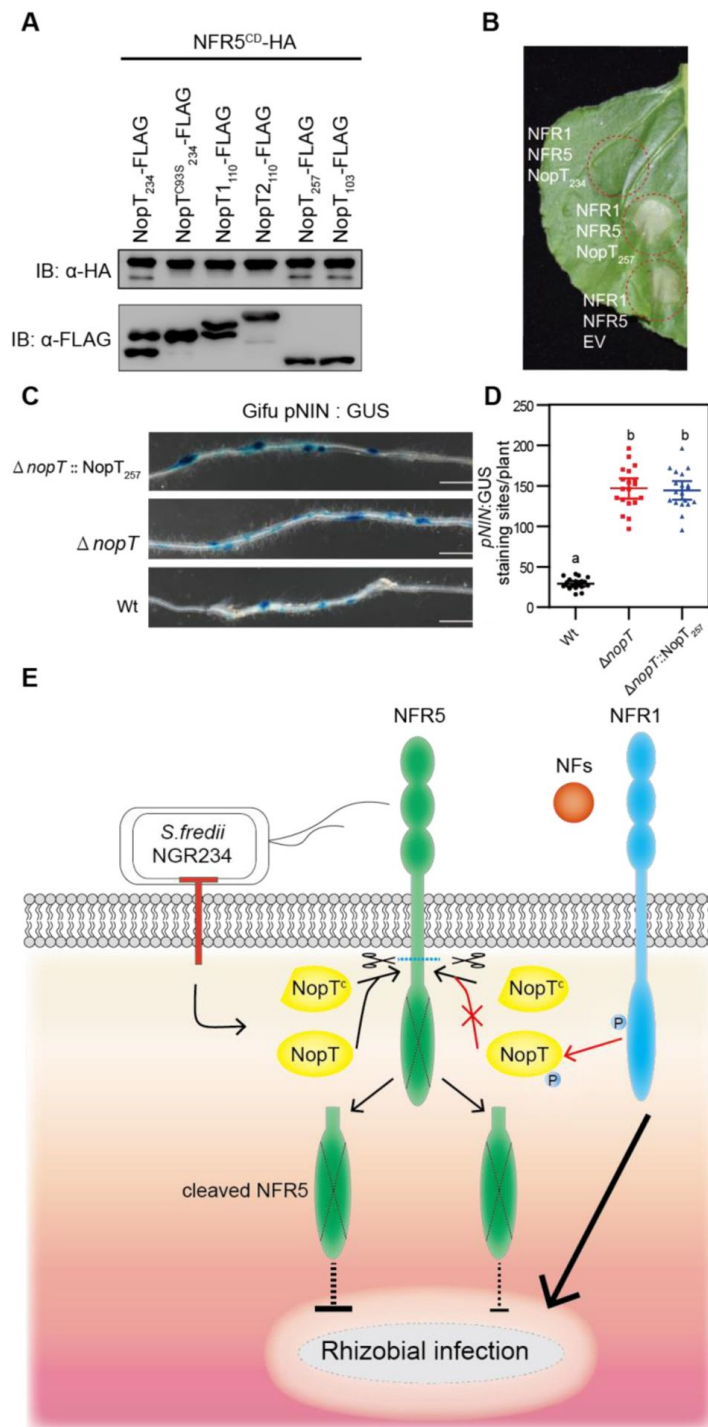


Figure 6.

Working model of NopT in association with NF receptors.

(A) NopT₂₃₄, NopT from *S. fredii* NGR234; NopT₁₁₀ and NopT₂₁₀, NopT homologs from *B. diazoefficiens* USDA110; NopT₂₅₇, NopT from *S. fredii* USDA257; NopT₁₀₃, NopT from *S. fredii* HH103. All NopT homologs were tested in the proteolysis assay using NFR5^{CD} as a target. (B) Rhizobial infection NopT₂₅₇ could not inhibit the cell death triggered by NFR1 and NFR5 in *N. benthamiana*. (C) The indicated rhizobial strains induced *nin::gus* expressing in *Lotus* root. Scar bars=2.5mm. (D) Statistical analysis of GUS staining in C (n=19, Student's *t*-test: *P* < 0.01). (E) A proposed model of NopT in association with NF receptors. Both NopT and NopT^c after autocleavage exert proteolytic activities to cleave NFR5 at the juxtamembrane domain to release the kinase domain of NFR5 (cleaved NFR5). NFR1 phosphorylates intact NopT, but not NopT^c, to block its proteinase activity.

(**Fig. S1A-C**), in which co-expression of two symbiotic receptors, NFR1 and NFR5, could lead to plant cell death to identify NopT, the only effector from *S. fredii* NGR234. *S. fredii* NGR234Δ*nopT* mutant strains hyperinfect *L. japonicus* roots, and overexpression of NopT in the roots reduces rhizobial infection (**Fig. 5A-5D**), consistent with the observed suppression of cell death in *N. benthamiana* leaves (**Fig. 1A**). The interaction between NopT and NFRs has been verified and is impaired by the acylation site of NopT (**Fig. 2** and S2A), suggesting that the plasma membrane localization of NopT is essential for targeting NFRs.

In the extensively studied context of microbial pathogenesis, various effectors from pathogens have been proposed to directly suppress the immune system and promote virulence in host cells (Tang et al., 2017; Schubert et al., 2020). The C58 protease family, including YopT and AvrPphB, represents such effectors that disarm the host defense system to favor pathogenic infection in both animals and plants (Shao et al., 2003a; Zhang et al., 2010). *Yersinia* YopT cleaves the Rho family GTPase to disrupt the actin cytoskeleton, inhibiting the phagocytosis of pathogens by hosts (Shao et al., 2003a).

Conversely, *P. syringae* AvrPphB cleaves multiple members of the RLCK (receptor-like cytoplasmic kinase) subfamily VII to suppress the PTI response (Zhang et al., 2010). In contrast, the cleavage of PBS1 functions as a “decoy” to trigger strong immunity against bacterial infection (Shao et al., 2003b). We confirmed that NopT cleaves NFR5 at the JM domain through different assays (**Fig. 3**). Mutants with five dispersed residues (S283, G294, Y303, A310, T311) at the JM domain (**Fig. 3F**) and the replacement of NFR5^{KD-CT} with NFP^{KD-CT} (**Fig. 3G**) resist cleavage by NopT, suggesting a joint decision of cleavage involving both the JM domain and KD-CT domain. Although belonging to the same protease family, the cleavage sites of NopT and YopT are polybasic (Shao et al., 2003a; Schmidt, 2011), in contrast to AvrPphB, whose cleavage site is canonical with seven adjacent amino acids involved (Kim et al., 2016). The different cleavage sites of YopT-type effectors indicate that bacteria regulate different physiological pathways to establish specific interactions with host plants.

Similar to the plant pathogenic effector AvrPphB, the protease activity of NopT appears to be conserved for this function by cleaving the soybean PBS1 homolog protein to restrict symbiosis (Khan et al., 2022). Indeed, transient expression of *S. fredii* NopT in *A. thaliana* and *N. tabacum* could induce strong immune responses and hypersensitive cell death (**Fig. S8**), indicating that rhizobial NopT might act as an avirulent effector in some nonhost plant species. Interestingly, we failed to obtain stable transgenic lines constitutively expressing NopT in *L. japonicus*. However, the generation of transgenic root hairs was not affected, implying that NopT might act as an Avr effector whose overexpression leads to whole plant lethality but not root growth. Therefore, given this conservation of protease function, it is consistent with a successful Avr effector that NopT has the ability to proteolyze not only NFR5 but also Arabidopsis AtLYK5 and *L. japonicus* LjLYS11, two receptors with chitin-binding affinities that mediate plant PTI responses (Cao et al., 2014; Gysel et al., 2021). NFR5, AtLYK5, and LjLYS11 are LYKs belonging to the same subfamily that lacks kinase activities. Thus, NopT may not only suppress the nod factor-mediated symbiotic signal to avoid excessive infection by *S. fredii* NGR234 but also function in inhibiting the signal triggered by chitin-like compounds. The cleavage of receptor-like kinases (RLKs) may be an overlooked strategy to regulate their function in plant biology compared to the well-studied phosphorylation to control RLK activation. While cleavage regulation is well studied in neurobiology, e.g., the Notch signal, the proteolysis of each Notch molecule leads to signaling activation (Kopan and Ilagan, 2009). Arabidopsis BAK1, identified as an universal co-receptor to regulate plant growth and MAMPs perception, undergoes proteolytic cleavage, which is important for both BR signaling and PTI response with an unclear mechanism (Zhou et al., 2019). NFR5 is a pseudokinase that functions as an output of NF signaling. The cleavage of NFR5 by rhizobial NopT releases the cytoplasmic domain of NFR5, but the precise function of whom is of interest to be determined.

The legume–rhizobial symbiosis is a unique case in the context of plant–microbe interactions. Mutual regulation between pathogenic effectors and host proteins has been reported in plant pathogenesis. For example, the *Pseudomonas* conserved effector AvrPtoB could directly target key components in the immunity pathway to promote its virulence (Goehre et al., 2008). However, the virulent function of AvrPtoB could either be enhanced or dampened by host proteins. AvrPtoB could be phosphorylated by SnRK2.8 to promote its virulence activity (Lei et al., 2020). Plant lectin receptor-like kinase LexRK-IX.2, Pto, and Fen kinases could also phosphorylate AvrPtoB to inactivate its E3 ligase activity and undermine its ability to suppress PTI (Ntoukakis et al., 2009; Xu et al., 2020). A similar strategy was used by legumes to dampen the “virulence” of the rhizobial effector NopT, favoring mutualistic infection. NFR1 phosphorylates NopT at several residues. The phosphomimetic NopTs (i.e., NopT^{T97D}, NopT^{S119D}, and NopT^{T257D}) are unable to cleave itself and NFR5 (Fig. 4C). The expression of these phosphomimetic NopTs separately in *S. fredii* NGR234Δnpt mutant strains cannot complement the hyperinfection phenotype (Fig. 5F). These data indicate that *L. japonicus* deploys NFR1 to phosphorylate NopT to dampen its catalytic activity and may protect NFR5 from cleavage, favoring rhizobial infections. However, NopT^{ΔN50}, mimic version of autocleaved NopT, retained the ability to interact with NFR5 but not NFR1 (Fig. 5D). Moreover, intact NopT, but not autocleaved NopT, are migrated by expressing NFR1^{CD} in immunoblotting (Fig. 3C and 4B), suggesting autocleaved NopT may escape from being phosphorylated and inactivated by NFR1.

In different rhizobium species, a variety of NopT homologs were characterized. NopT homologs from some *Bradyrhizobium* species lose the ability to cleave NFR5, while NopT homologs from some *Sinorhizobium*, truncated versions without about 50 a.a. at its N-terminus, retain the ability to cleave NFR5 (Fig. 6A). The type three secretion signal depends on the N-terminal of effectors (Akedo and Galán, 2005). However, an earlier study has shown that NopT from *S. fredii* HH103 exhibited a positive effector on nodulation in soybean SN14 cultivar (Li et al., 2023), suggesting that another unknown mechanism might be involved in NopT^{HH103} secretion. Interestingly, these truncated

NopTs still keep the function to cleave NFR5 (Fig. 6A) and autocleaved NopT_{NGR234} cannot interact with NFR1 (Fig. 5D), indicating that NopTs from those strains and autocleaved NopT_{NGR234} might lose the ability to be suppressed by NFR1. Notably, natural truncated NopT from *S. fredii* USDA257 and *S. fredii* HH103 are the same in protein sequence, but they lack the acylation site compared with NopT_{NGR234}. Therefore, NopT_{USDA257} and NopT_{HH103} lose the ability to suppress cell death triggered by NFR1 and NFR5 in *N. benthamiana* leaves (Fig. 6B) and failed to complement the massive infection of NGR234Δnpt (Fig. 6C and 6D), implying NopT_{USDA257} and NopT_{HH103} may have a different target from NopT_{NGR234} in the host plant. In addition, knockout of NopT in *Bradyrhizobium* sp. ORS3257 was shown to result in uninfected nodule formation, suggesting that, in this system, NopT_{ORS3257} might play a role in nodule formation (Teulet et al., 2019). The versatile function of NopT homologs in different rhizobium species suggests that different regulatory processes might be involved in specific rhizobial symbiosis.

In this study, we present a mutual regulation model in which the rhizobial effector NopT directly targets NFR5 to suppress symbiosis signals and is then inactivated by NFR1 via phosphorylation. NopT appears to retain some pathogenic-like features in that it can cleave the *Arabidopsis* chitin receptor AtLYK5 and induce cell death in some non-leguminous plants, such as *N. tabacum* and *Arabidopsis*. Thus, the cleavage of NFR5 by NopT might be due to the structure similarity to AtLYK5 targeted by NopT. The suppression of proteolytic activity of NopT via phosphorylation by NFR1 appears to work as a feedback loop to allow symbiotic signaling in compatible hosts. The model (Fig. 6E) provides essential insights into the legume–rhizobial symbiosis, where rhizobia deploy an effector to dampen symbiotic signaling, and legumes slightly recur infection by suppressing NopT to employ the nod factor signal in response. Even though this couple, legume and rhizobia, tries to get together and strives to improve their living conditions, we catch a glimpse into the arguments and the recovery conversations between them.

Materials and Methods

Plant materials

Lotus japonicus Gifu B-129 cultivar was provided by the Center for Carbohydrate Recognition and Signaling (<https://lotus.au.dk/>), and used as wild type for nodulation assays. The Gifu *pNIN:GUS* transgenic line was kindly provided by Dr. J. Stougaard from the Aarhus University (Heckmann et al., 2011). All *L. japonicus* seeds were treated with concentrated sulfuric acid for 10 min, followed by surface sterilization in 1% (w/v) NaClO for 8 min. After synchronization of germination at 4°C in the dark for 2 d, the seeds were placed on half-strength Murashige & Skoog (MS) medium containing 0.8% (w/v) agar for germination at 22°C for 2 d in the dark. Seedlings were then transferred to growth pots containing vermiculite supplied with half-strength B&D (Broughton & Dilworth) medium without nitrogen under long-day conditions (16-h light/8-h dark) at 22°C. Ten-day-old seedlings were inoculated with *Sinorhizobium fredii* NGR234 (OD₆₀₀=0.02) for infection assays.

Plasmid construction

All plasmids used in this study were generated using MultiF Seamless Assembly Mix (ABclonal, catalog number RK21020). The plasmid pGWB514 was used as an original plasmid for all binary vector construction (Nakagawa et al., 2007). Briefly, Different tags including HA tag, Myc tag, FLAG tag, Strep tag, nLUC, cLUC, and GFP were respectively amplified using a forward primer containing a *KpnI* site and ligated into pGWB514 digested with *XbaI* and *SacI* using seamless cloning method as described above. The generated plasmids were named pG5XX-HA, pG5XX-Myc, pG5XX-FLAG, pG5XX-nLUC, pG5XX-cLUC, pG5XX-GFP, respectively. The coding sequences of *L. japonicus* *NFR1* and *NFR5* were cloned into pGWB514 and pG517 between *XbaI/KpnI* to generate *NFR1*-HA and *NFR5*-Myc fusion constructs under the control of the 35S promoter. then, the DNA fragment containing *pro35S:NFR5*-Myc-NosT was amplified and cloned into pG514-*NFR1* at *SbfI*. The final plasmids contained *pro35S:NFR1*-HA- NosT and *pro35S:NFR5*-Myc-NosT. Rhizobial genes encoding effectors from *S. fredii* NGR234 were cloned into pG5xx-FLAG between *XbaI/KpnI*. To detect the interaction between *NFR1/NFR5* and *NopT* in *N. benthamiana*, the full-length *NFR1/NFR5*, *NopT* and *AtFLS2* coding sequences were cloned into pG5xx-Nluc, pG5xx-Nluc, pSPYCE(MR) and pSPYNE(R)173 (Waadt et al., 2008). For cleavage assay in *N. benthamiana*, the *NFR5* and *NopT* coding sequences were cloned into pG5XX-FLAG GFP and pG5xx-Strep respectively. For cleavage assay in *E. coli*, the sequence encoding the cytoplasmic domain of *NFR5* was cloned into pACYC duet Strep-HA, the sequences encoding the kinase domain of *NFR5* fused to HA tag and the juxtamembrane domain of *NFR5* fused to the GFP tag was cloned into pSUMO. The point mutants in *NFR5*^{CD} and *NopT* were created from the pACYC duet Strep-*NFR5*^{CD}-HA and pET28a *NopT*-FLAG template using site-directed mutagenesis PCR. For *in vivo* cleavage assay in *L. japonicus*, *NopT*-FLAG or *NopT*^{C93S}-FLAG was cloned into pUB GFP between *XbaI* and *KpnI*, and then 35S-*NFR5*-Myc-NosT was cloned in pUB *NopT* GFP at *PstI*. For phosphorylation assay, the sequence encoding the cytoplasmic domain of *NFR1* fused to a Myc tag was cloned into pCDF duet. For rhizobium infection assays, 300 bp upstream of *NopT* and site mutants of *NopT* were fused by overlapping PCR and cloned into pH60.

Transient expressing and in *Nicotiana* species

The plasmids for transient expression of INF1, Avr3 and R3a in *N. benthamiana* were kindly gifted by Dr. Juan Du from HZAU. *Agrobacterium* (*Agrobacterium tumefaciens*) strain EHA105 carrying various constructs were mixed with an *Agrobacterium* culture harboring the P19 suppressor in an infiltration buffer (10 mM MgCl₂, 10 mM MES-KOH pH 5.8 and 200 μM acetosyringone). After incubating 2 h at room temperature, *Agrobacterium* cultures were infiltrated into the leaves of 4–

week-old *N. benthamiana* and *N. tabacum*. Leaves were harvested for analysis at two days post infiltration. The cell death suppression assay in *N. benthamiana* was performed as described previously (Wang et al., 2011 [DOI](#)).

Hairy Root Transformation in *Lotus japonicus*

The hairy root transformation assays were performed as described previously (Li et al., 2018 [DOI](#)). In brief, surface-sterilized seeds were germinated and grown in the MS plate without sucrose (23°C dark for first 3 days and 23°C 16 h light/8 h dark for next two days). Then seeds were cut at the middle of the hypocotyl and co-cultivation with *A. rhizogenes* carry pUB GFP and pUB NopT-FLAG GFP respectively (23°C dark for first 3 day and 23°C 16 h light/8 h dark for next 2 day). Transfer the plants onto B5 media. After about 15 days, the plants were picked up for further analysis by cutting the root without expressing florescent proteins. For rhizobial infection assays, data were collected at 9dpi for *S.fredii* NGR234 and 6dpi for MAFF303099

NFR5 cytoplasmic domain cleavage assay

For cleavage assay in *E. coli*, Strep-NFR5^{CD}-HA or Strep-NFR5^{CD}-HA were co-expressed with NopT-FLAG or NopT^{C93S}-FLAG by using Duet expressing vectors (Novagen) (Han et al., 2017 [DOI](#)). Bacterial cell cultures were treated with 0.5 mM IPTG at 22°C for 15 h. The bacterial pellet was resuspended in extraction buffer consisting of Tris-HCl (pH 7.3), 150 mM NaCl, 5 mM EDTA, 0.5% (w/v) SDS and 0.5% (v/v) Triton X-100. After adding SDS loading buffer and boiling at 100°C for 5 min, total proteins were detected with anti-HA-peroxidase (Sigma, clone 3F10), anti-FLAG (Sigma, F1804), and anti-Strep (ABclonal, AB_2863792) antibodies.

For *in vitro* cleavage assays, Sumo-NFR5^{JM}-GFP and NopT-FLAG recombinant proteins were purified, and 3 µg NopT-FLAG and 1 µg Sumo-NFR5^{JM}-GFP were incubated in a test tube at 22°C for 7 h. The cleavage products were detected with anti-GFP antibody by immunoblot (ABclonal, AB_2770402).

For *in vivo* cleavage assays, the transgenic root was collected at about 15 days after induction of hairy roots. 2% protease inhibitors (Sigma, P9599) were added in extraction buffer. The actin proteins were detected with anti-Actin antibody (ABclonal, AC009).

Phosphorylation assays

For phosphorylation assays in *E. coli*, His-NFR1^{CD}-Myc or His-NFR1^{CD-K351E}-Myc and NopT-FLAG-His were co-expressed by using Duet expressing vectors (Novagen) with 0.5 mM IPTG at 28°C for 18 h. After purifying NFR1^{CD} and NopT at the same time, CIAP (TaKaRa, 2250B) was added and incubated at 37°C for 3 h. The mobility shift of NFR1^{CD} and NopT was detected with anti-Myc (Bio-Legend, 626808) and anti-FLAG (Sigma, F1804) antibodies by immunoblot analysis.

For phosphorylation assays in *L. Japonicus* roots, NopT^{C93S}-FLAG was expressed in the *nfr1* mutant and wild type plants, respectively. Transfer the plant with florescent root to half-strength B&D media for 5 days. After treat with water and rhizobia respectively, the total proteins were drawn by extraction buffer with addition 2% protease inhibitors cocktail (Sigma, P9599) and 2% phosphatase inhibitors cocktail (Yeasen, 20109-A), then followed with TCA precipitation to remove various contaminants (e.g. EDTA, surfactant). Phosphorylation of NopT^{C93S} proteins was analyzed by 50µM Zn²⁺-Phos-tag SDS-PAGE and detected with anti-FLAG (Sigma, F1804) antibodies by immunoblot analysis.

BiFC and Split-LUC assays in *N. benthamiana*

In BiFC assay, the eYFP fluorescence of fusion proteins was assayed 2–3 days after infiltration using a confocal microscope (Leica TCS SP8). For the Split-LUC assay, *N. benthamiana* leaves were collected at 2 dpi sprayed with a solution of 1 mM D-luciferin (Promega, E1603) and 0.02% (v/v)

Triton X-100. After a dark-adaptation of 5 min, bioluminescence images were acquired using a Tanon Bio-Imaging System (Tanon, Shanghai, 4600).

Liquid chromatography mass spectrometry

For proteomics analysis of NopT phosphorylation sites, *NopT* and *NFR1-CD* or *NFR1-CD^{K351E}* were co-expressed in *E. coli*, then proteins were purified using Ni-Charged Resin (GenScript, Cat. No. L00223). The bands of NopT^P and NopT, as control, were cut and destained by 50 mM triethylammonium bicarbonate (TEAB) and 50% (v/v) acetonitrile (ACN) in 50 mM TEAB followed by washing with 100% ACN until the gel turned white. The protein bands were then in-gel digested with sequencing-grade trypsin. LC-MS analysis was performed by Novogene Co., Ltd. (Beijing, China).

For identification of the cleavage site on NFR5, NopT and the recombinant protein, His-Sumo-NFR5-GST, were co-expressed in *E. coli* followed by protein purification using Glutathione Resin (GenScript, Cat. No. L00206). About 80µg cleaved NFR5 was collected for analysis. The protein bands were digested with trypsin, glu-C, chymotrypsin and pepsin. LC-MS analysis was performed by Oulu Biotechnology Co., Ltd. (Shanghai, China).

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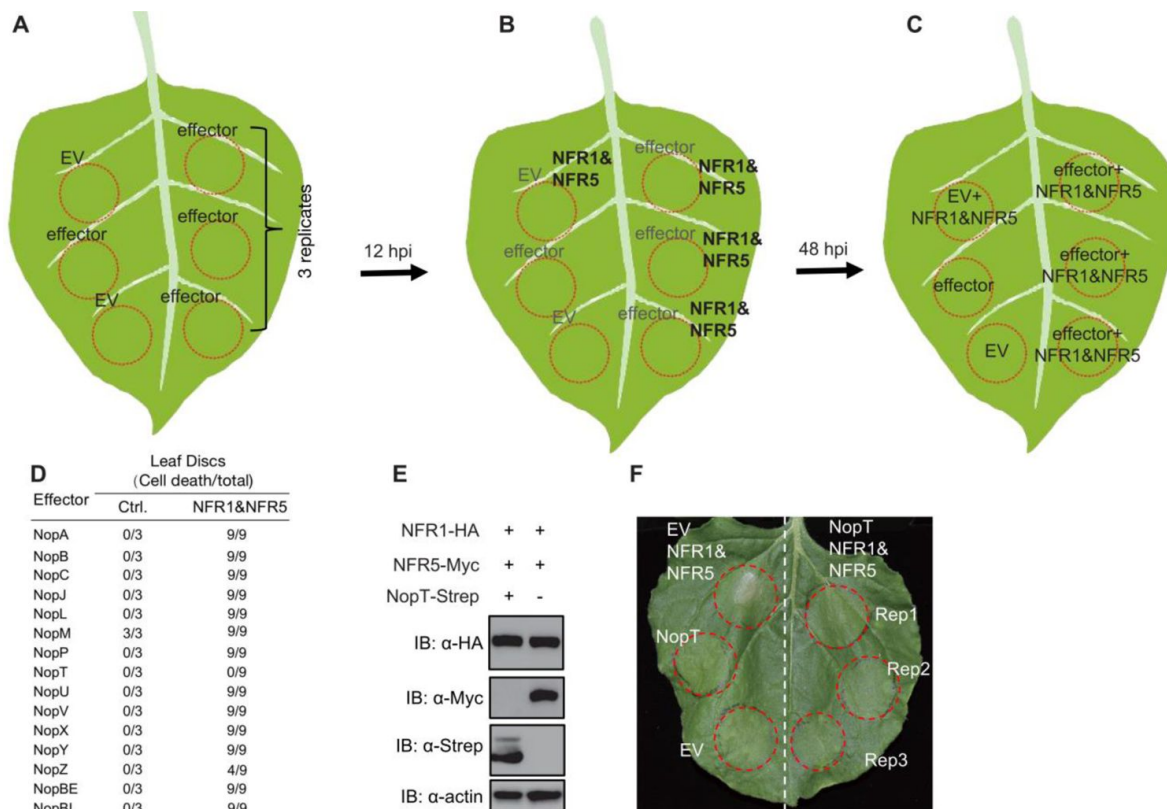
Author Contributions

H.B., Y.C., and G.S. conceived the idea and designed the research, H.B., Y.W., Y.L., and Y.Y. performed experiments; H.B., Y.W., H. L., H.Z., and Y.C. analyzed the data, Q.W and S.X identified the cleavage site by MS. H.B., Y.C., and G.S. wrote the paper.

Supplemental Figures

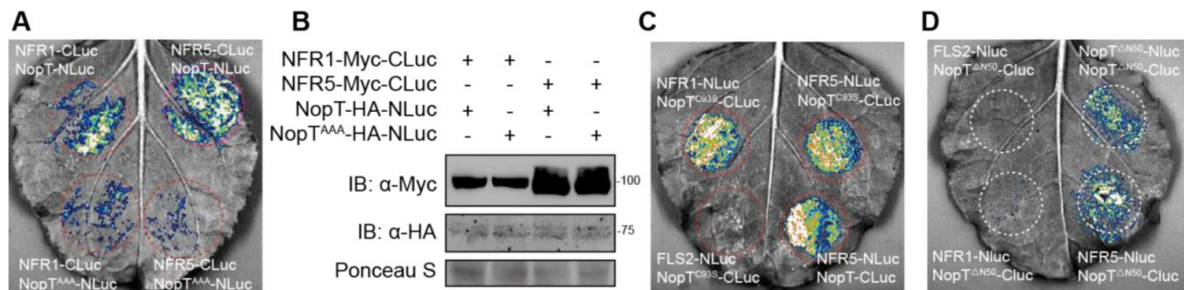
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Akeda Y., Galán J.E (2005) **Chaperone release and unfolding of substrates in type III secretion** *Nature* **437**:911–915



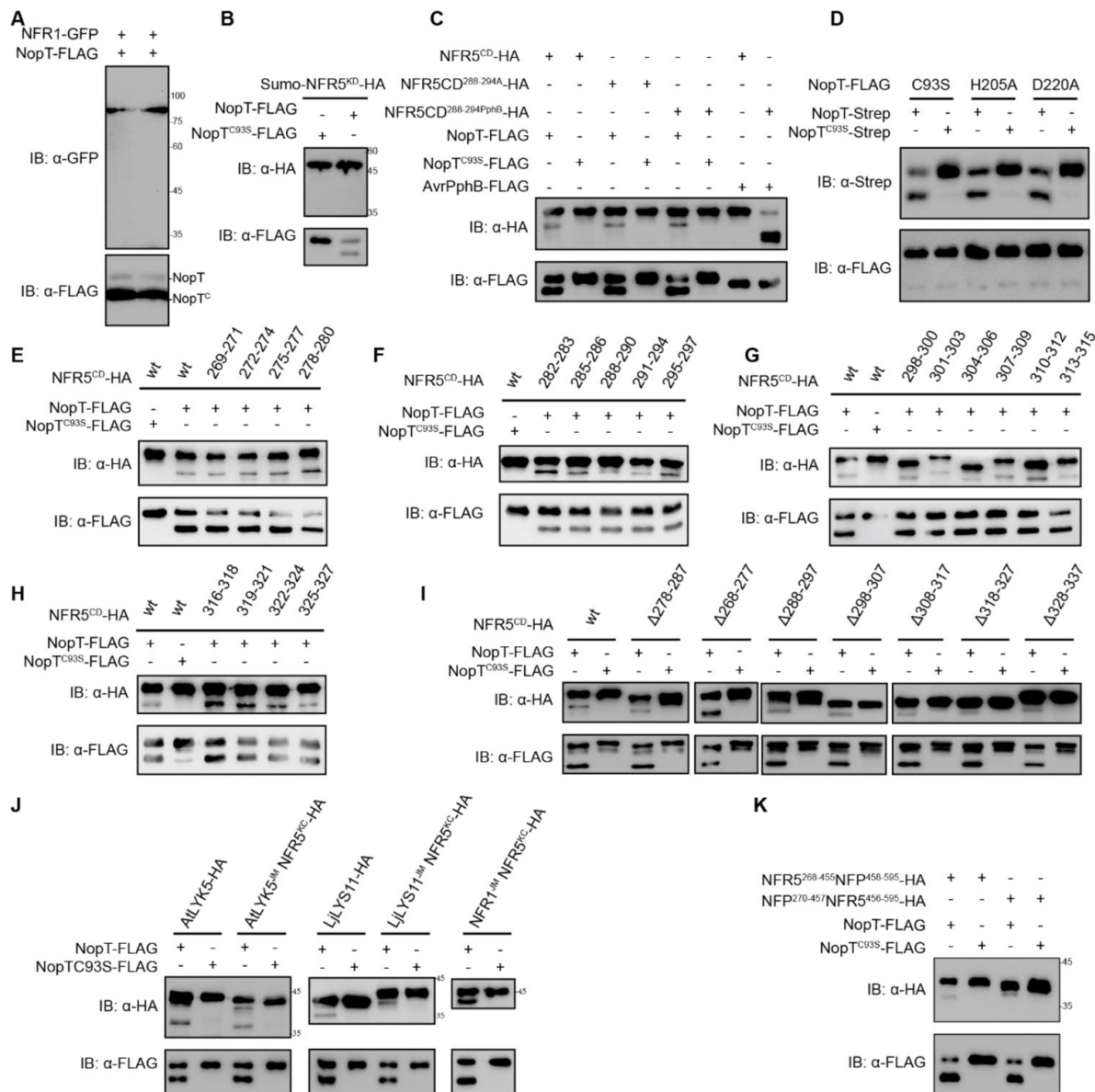
Supplemental Fig. 1.

Analyses of the functions of all 15 effectors from *S. fredii* NGR234 in preventing LjNFR1- and LjNFR5-induced cell death in *N. benthamiana* leaves. 15 genes from *S. fredii* NGR234 encoding effector proteins were amplified and cloned into binary vector for expression of Strep-tagged proteins in *N. benthamiana*. LjNFR1 and LjNFR5 from *L. japonicus* were C-terminally tagged with HA and Myc, respectively, and cloned into binary vectors for expression in *N. benthamiana*. All plasmids were electroporated into *Agrobacterium tumefaciens* EH105. (A) *Agrobacterium* strain harboring each effector gene or control empty vector (EV) were infiltrated into *N. benthamiana*. 12 hour post infiltration (hpi), *agrobacterium* strains harboring LjNFR1 and LjNFR5 were mixed and hand-infiltrated in the same leaf discs as shown in (B). 48 hpi, cell death phenotypes were evaluated (C). At least three replicates in the same leaf were performed to test the function of each effector in suppressing cell death-triggered by LjNFR1 and LjNFR5. (D) Results of all 15 effector proteins in suppressing programmed cell death by overexpression of NFR1 and NFR5. (E) Abundance of LjNFR1, LjNFR5, and NopT proteins, as measured by immunoblot with specific antibodies. Leaf discs expressing LjNFR1, LjNFR5, and NopT or LjNFR1 and LjNFR5 and with EV were used for immunoblot analyses. Actin was used as loading control. (F) Another example of suppression of programmed cell death by NopT in the leaf discs coexpressing NFR1 and NFR5. Rep1, Rep2, and Rep3 represent three technical repeats in the same leaf.



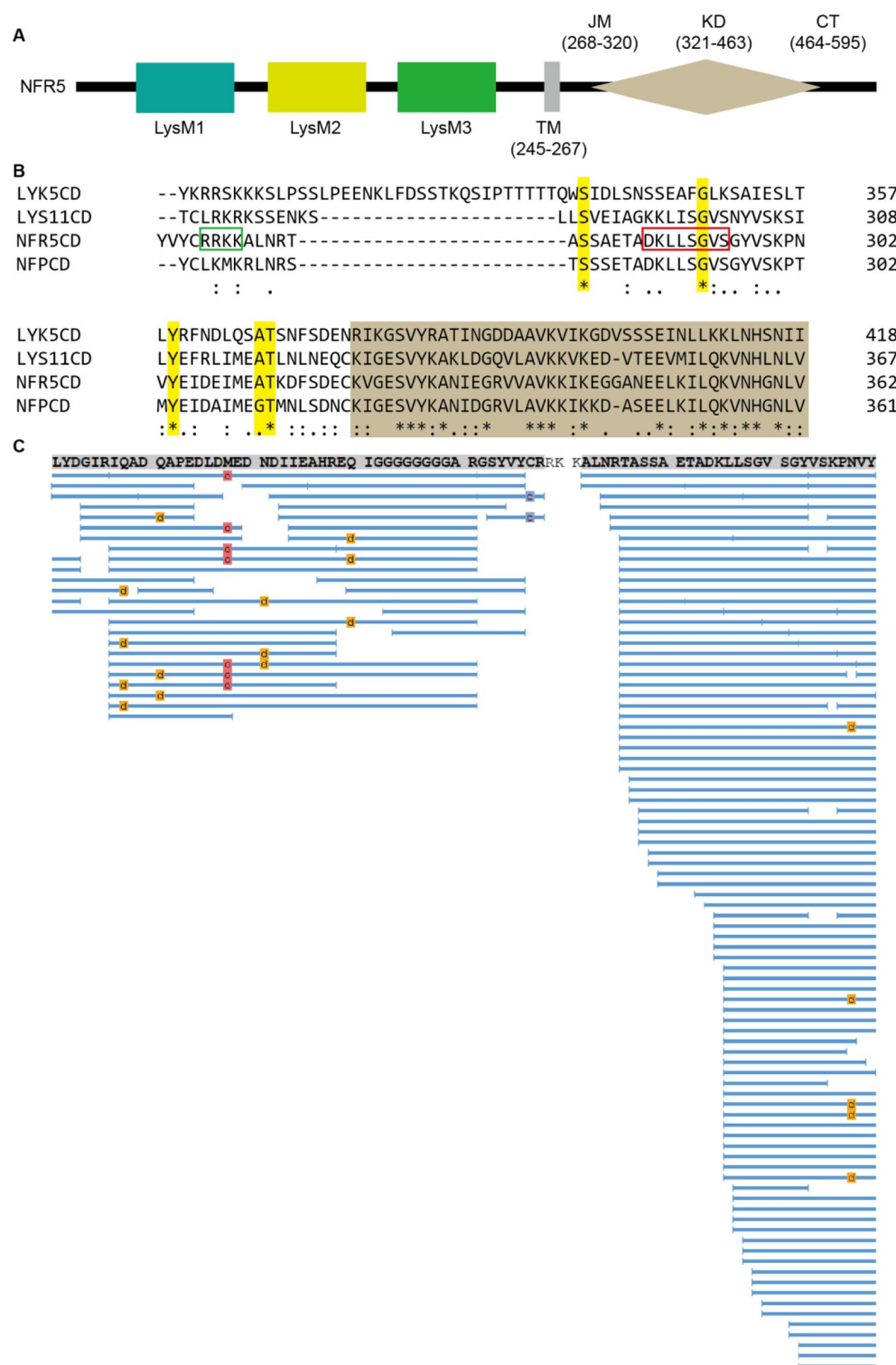
Supplemental Fig. 2.

Split-LUC assay testing the interactions between different NopT mutants and NFRs. (A) The interaction between NopT with mutated acylation site and NFR1/NFR5. NopT^{AAA} (G50A C51A C52A). (B) The protein expressing of indicating genes in the leave of (A). (C) Nop^{C93S} interact with NFR1 and NFR5. (D) The interaction between NopT^{ΔN50} and NFR5. NopT^{ΔN50} represents a truncated NopT with deletion of about 50 a.a. from its N-terminus.



Supplemental Fig. 3.

NopT cleaves NFR5 at the juxtamembrane domain. (A) NopT cannot cleave NFR1-GFP expressed in *N. benthamiana* leaves. recombinant Sumo-NFR5^{KD}-HA lacking the juxtamembrane (JM) domain. (B) NopT could not cleave Sumo-NFR5^{KD}-HA, a recombinant protein expressed in *E. coli* with the juxtamembrane domain of NFR5 was replaced with Sumo tag. (C) Cleavage assay using two mutant versions of NFR5^{CD} in the presence of NopT or AvrPphB. NFR5^{CD288-294A} represents a mutant NFR5^{CD} with residues 288 to 294 replaced with seven alanines; NFR5^{CD288-294PphB} represents a mutant NFR5^{CD} with residues 288 to 294 replaced with AvrPphB recognition sites. (D) Autocleavage assay, using different NopT variants with single amino acid mutations. H205A and D220A indicate His-205 and Asp-220 replaced with alanine, respectively. (E-H) NopT cleavage assay using 19 variants of NFR5^{CD} with three adjacent amino acids replaced with three alanines. The numbers indicate the location of the three residues replaced with alanines. (I) NopT cleavage assay using seven truncated variants of NFR5^{CD} with 10 adjacent amino acids deleted. The numbers indicate the location of the 10 deleted residues. (J) NopT but not NopT^{C93S} could cleave the CDs of AtLYK5 and LjLYS11 expressed in *E. coli*. NopT could cleave both AtLYK5^{JM}-NFR5^{KC}, LjLYS11^{JM}-NFR5^{KC} and LjNFR1^{JM}-NFR5^{KC}, three recombinant proteins expressed in *E. coli* with the JM of NFR5 was replaced with the JMs from AtLYK5, LjLYS11 and NFR1, respectively. (K) NopT but not NopT^{C93S} could cleave recombinant protein NFR5268-445-NFP456-595 and NFP270-457-NFR5456-595.



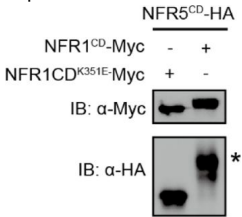
Supplemental Fig. 4.

Characterization of cleavage site of LysM receptor. (A) Domain and motif annotation of NFR5. (B) Protein sequence alignment of the cytoplasmic domain of AtLYK5, LjLYS11, LjNFR5 and MtNFP. Yellow boxes delineated highly conserved residues (S283, G294, Y303, A310, T311) and brown box delineated kinase domain. Green frame delineated cleavage site of NFR5 and red frame delineated residues similar to NopT autocleavage region. (C) Mass spectrometry analysis of cleaved NFR5. Blue lines indicated the peptides characterized by MS.

Supplemental Fig. 5.

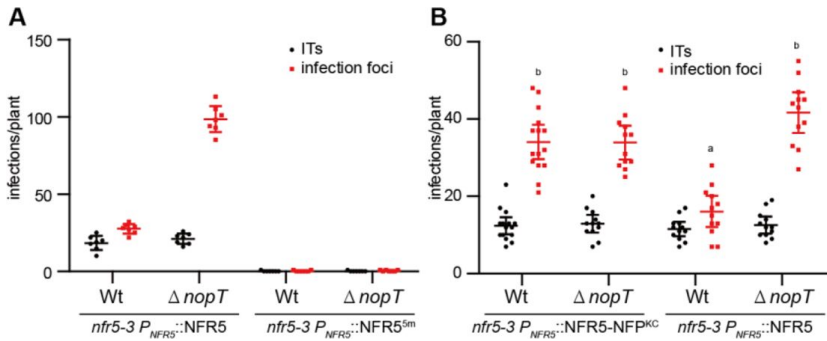
The cytoplasmic domains (CDs) of NFR1, NFR5, and kinase-dead NFR1CD^{K351E} were used for kinase assays in *E. coli*.

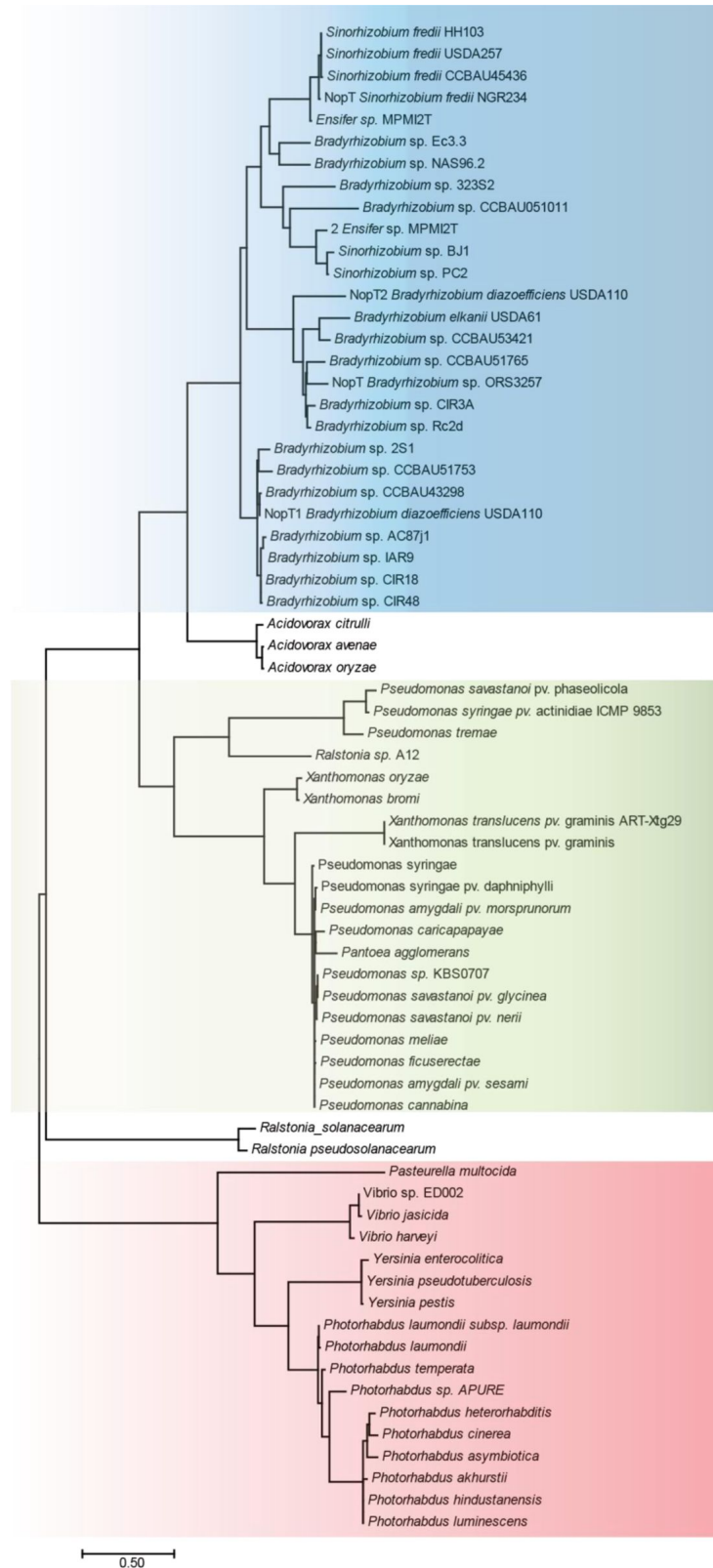
Proteins were detected by immunoblotting using anti-HA and anti-Myc antibodies. Asterisk indicated the band retardation on the gel representing the phosphorylated NFR5^{CD} proteins.



Supplemental Fig. 6.

Rhizobial infection in the mutant versions of NFR5. (A) NFR5^{5m} failed to form rhizobial infection (n=7, Student's t-test: P < 0.01). (B) NFR5-NFP^{KD-CT} could rescue the rhizobial infection of wild type NGR234 (n=10, Student's t-test: P < 0.01). The transgenic roots expressing NFR5 or NFR5^{5m} or NFR5-NFP^{KCT} in *nfr5-3* background under the control of native promoter were inoculated with GFP-labelled wild type NGR234 and NGR234Δ*nopT* respectively. The data was collected 9 days post inoculation.



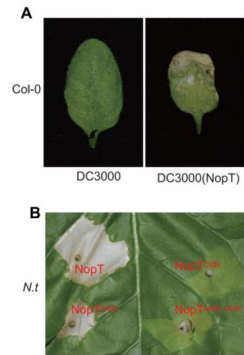


Supplemental Fig. 7.

Phylogenetic tree based on the amino acid sequence of NopT homologs from different bacterial species.

Supplemental Fig. 8.

Transient expression of NopT triggers cell death in *Arabidopsis thaliana* and *Nicotiana tabacum*. (A) *Pseudomonas syringae* pv tomato DC3000 harboring control plasmids (left) or expressing NopT (right) were infiltrated into *Arabidopsis* leaves. Note the cell death in the right-side panel. (B) *Agrobacterium* strains harboring *NopT*, *NopT* with an N-terminal 50-amino acid deletion, *NopT*^{C93S}, or *NopT*^{C93S} with an N-terminal 50-amino acid deletion were infiltrated into *N. tabacum* leaves.



Supplemental Table 1.

Phosphopeptides identified by liquid chromatography– mass spectrometry.

An *in vitro* kinase assay was performed using the CD of NFR1 and NopT. NopT was separated by SDS-PAGE, and the digested gel slices representing the phosphorylated NopT were used for analysis. The deduced phosphopeptides are listed.

Sequence	Modifications	SITE	PSMs NFR1KE+NopT	PSMs NFR1+NopT
ETLANVELRTK	1xPhospho [T10]	37T	-	1
ETLANVELR	1xPhospho [T2]	29T	-	1
ETLANVELRTKSPSAECPDK	1xCarbamidomethyl [C17]; 1xPhospho [T10]	37T	-	1
TKSPSAECPDK	1xCarbamidomethyl [C8]; 1xPhospho [S5]	41S	-	1
SPSAECPDK	1xCarbamidomethyl [C6]; 1xPhospho [S3]	41S	-	1
SPSAECPDKMGCCASKPQASDPNPNPSTSSPARPSTSLFR	3xCarbamidomethyl [C6; C12; C13]; 1xOxidation [M10]		1	1
SPSAECPDKMGCCASKPQASDPNPNPSTSSPARPSTSLFR	3xCarbamidomethyl [C6; C12; C13]; 1xPhospho [S/T]; 1xOxidation [M10]		-	4
MGCCASKPQASDPNPNPSTSSPARPSTSLFR	2xCarbamidomethyl [C3; C4]; 1xPhospho [T/S]; 1xOxidation [M1]		-	11
MGCCASKPQASDPNPNPSTSSPARPSTSLFR	2xCarbamidomethyl [C3; C4]; 1xPhospho [T/S]		-	9
MGCCASKPQASDPNPNPSTSSPARPSTSLFR	2xCarbamidomethyl [C3; C4]; 2xPhospho [T/S]	65S 66T	-	4
MGCCASKPQASDPNPNPSTSSPARPSTSLFR	2xCarbamidomethyl [C3; C4]; 1xOxidation [M1]		4	2
MGCCASKPQASDPNPNPSTSSPARPSTSLFR	2xCarbamidomethyl [C3; C4]; 2xPhospho [T/S]; 1xOxidation [M1]		-	4
TAELAQANADGICVGLTAEWLR	1xCarbamidomethyl [C13]; 1xPhospho [T17]	97T	-	10
NLNSHPSIR	1xPhospho [S7]	109S	-	2
NLNSHPSIRMEALVPGSQR	1xPhospho [S7]; 1xOxidation [M10]	119S	-	1
MEALVPGSQR	1xPhospho [S8]	119S	-	1
MEALVPGSQR	1xPhospho [S8]; 1xOxidation [M1]	119S	-	1
RQGAGPSEADFAAQNTMLQK	1xPhospho [T/S]; 1xOxidation [M17]		-	3
RQGAGPSEADFAAQNTMLQK	1xPhospho [T16]	156T	-	2
QGAGPSEADFAAQNTMLQK	1xPhospho [T/S]; 1xOxidation [M16]		-	3
QGAGPSEADFAAQNTMLQK	1xPhospho [T15]	156T	-	2
SLANRYSNPNR	1xPhospho [S7]	245S	-	2
QHLTVITQK	1xPhospho [T8]	257T	-	1

Used for	Target genes	Names	Sequences (from 5' to 3')	
Overexpression in <i>Nicotiana benthamiana</i>	NopT or NopT ^{C93S}	p5X-NopT-F p5X-NopT-R	agaacacgggggactctagaatgcacagtcacatcag ccgctgttatcgggtacctgtcatcttttgggtggtcac	
	NopT ^{ΔN50}	p5X-NopTN50-F	agaacacgggggactctagaatgtgctgcgccagcaag	
	NFR1	p5X-NFR1-F p5X-NFR1-R	agaacacgggggactctagaatgaagctaaaaactggtctact ccgctgttatcgggtacctcttctcacagac	
	NFR5	p5X-NFR5-F p5X-NFR5-R p5X-NFR5gfp-R	agaacacgggggactctagaatggctgtcttcttcttacc ccgctgttatcgggtaccacgtgcagtaat tcccgggagcgggtaccacgtgcagtaatggaagt	
	AvrPphB	p5X-AvrPphB-F p5X-AvrPphB-R	agaacacgggggactctagaatgaaaataggtacgcaggcc ccgctgttatcgggtaccgaaactctaaactcgtttacgc	
templates for protein expression	NopT _{USDA257}	p5X-257T-F p5X-257T-R	agaacacgggggactctagatgctgctgatccgaacaat ccgctgttatcgggtacctgtcaccttttgggtggtcaccg	
	NopT1 _{USDA110}	p5X-110T1-F p5X-110T1-R	agaacacgggggactctagaatgtatgatcgaatcggtgg ccgctgttatcgggtaccctgcatcctttgctgcg	
	NopT2 _{USDA110}	p5X-110T2-F p5X-110T2-R	agaacacgggggactctagaatgtataatcagtcgatggc ccgctgttatcgggtacccgatgaggttcgc	
Protein Expression in <i>E.coli</i>	NopT or NopT ^{C93S}	28a NopT-F 28a FLAG-R 28a Strep-R	tgggtcgcggatccgaattcatgcacagtcacatcag tcgagtgccggccgaagcttctgtcatcgtcatcctgttag tcgagtgccggccgaagctttttcaaatgaggatgagaccatcc	
		AvrPphB	ET28a-pphb-F	tgggtcgcggatccgaattcaaaataggtacgcagccac
		NopT _{USDA257}	ET28a-257T-F	tgggtcgcggatccgaattcctcgtgatccgaacaa
	NopT1 _{USDA110}	ET28a-110T1-F	tgggtcgcggatccgaattctatgatcgaatcggtgctc	
	NopT2 _{USDA110}	ET28a-110T2-F	tgggtcgcggatccgaattctataatcagtcgatggcgaatac	
	NFR1CD	duet NFR1-F duet Sac1nost-R	accacagccagatccgagataccagaagaagaagaagaagc ggcgcgccgagctcgcgatcgggaaattcgagct	
		MAL-NFR1-F MAL-NFR1-R	ggaaggatttcagaattcggatccagataccagaagaagaagaag cgacggccagtgccaagctttcatctctcacagacagtaattatgag	
		NFR5CD	ACSH-NFR5-F ACSH-NFR5-R	agaagcgcggatccgaattctatgtatactccgcagaagaag cgtatgggtagcttccaagcttacgtgcagtaatggaagtc
	MAL-NFR5-F MAL-NFR5-R		ggaaggatttcagaattcggatcctatgtatactccgcagaagaag cgacggccagtgccaagctttcaacgtgcagtaatggaagtc	
	NFR5KD		SUMO 5KD-F SUMO HA-R	ccgcgaacagattggttaaggttgggaatcagtgatc gctcgaattcggatcctttcacgcatagtcaggaacatcg
	NFR5JM	SUMO 5JM-F SUMO 5JM-R	ccgcgaacagattggtatgtatactccgcagaagaag gctcgaattcggatcctttcacgcatagtcaggaacatcg	
	sumo-NFR5KD-HA	duet sumo-F duet HA-R	catcaccacagccagatgggtccctgcaggac cattatgcggccgaagcttttacgcatagtcaggaacatcg	
	sumo-NFR5JM-GFP	duet sumo-F duet GFP-R	catcaccacagccagatgggtccctgcaggac atgcggccgaagcttttactgtacaactcatccatacc	

Supplemental Table 2.

Primers used in this study.

	AtLYK5	ACSH-LYK5-F ACSH-LYK5-R	gcgcggatccgaattctacaacgaaggtctaagaag ggtagcttccaagctgttccaagagagccg
	LjLYS11	ACSH-LYS11-F ACSH-LYS11-R	agaagcgcggatccgaattcactgtctgaggaaagaaag cgtatgggtagcttccaagcttacgagctgctatcaaagt
	LYK5JM NFR5KD	LYK5-NFR5-F NFR5 LYK5-R	cgaaaacagaaggttggggaatcagt ccccaacctttctgtttcgtcgtgaaatt
	LYS11JM NFR5KD	LYS11 NFR5-F NFR5 LYS11R	gagcagtgtgaagttggggaatcag ccccaaccttacactgctcattgagg
site mutations construction	NopT	T93S-F T93S-F	gaatctcctcggcttactgc gacggagattccatctcgttc
		97D-F 97D-R	ggcttggatgcggagtgctgcg ctccgatccaagccgacgacg
		97A-F 97A-R	ggcttggctgcggagtgctgc ctccgagccaagccgacg
		109D-F 109D-R	catccgatatccgaatggaggccctag ttcgatatccggatgactgttaggttacgc
		109A-F 109A-R	catccggcaatccgaatggag ttcgattgccggatgactgttg
		119D-R 119A-F	cccggaatcaaaagcacgcctcagc gcctttgatctccgggtactagggcc
		119A-F 119A-R	cccggaagcgaaggcac ctttgcgctccgggtactagg
		156D-F 156D-R	gcaaaacgatattgtcagaaagcaggc gcaacatatcgttttcgcccgaagtcggcc
		156A-F 156A-R	gcaaaacgctattgtcagaaagcag ctgcaacatagcgttttcgcccgaagtc
		H205A-F H205A-R	ggcggcaacacgttgcgacc caacgggtgttgcgcccctcagc
		D220A-F D220A-R	ctcttcgtcctaatttcgccaatttac gccgaaattaggagcgaagagcgtgtgtttc
		245D-F 245D-R	cgtacgacaatcaaaccgacg ttggattgtctagcgattggctaggct
		257 mutants	changing the sequence of R primer at nopT ₂₅₇ when cloned into vectors
	NFR1	K351E-F K351E-R	gcaattgagaagatggatgtacaagcatc catccattcttcaattgctgtttcttgc
	NFR5	269-271A-F 269-271A-R	ctatgcagctgcacgcagaaagaaggctctg gcgtgcagctgcataagaatcgatccgcgc
		272-274A-F 272-274A-R	ctgcgtgcagctaaggctctgaataggactgc cttagctgcagcgcagtatacatagaattcgatcc

Supplemental Table 2. (continued)

	275-277A-F	gaaaggcagctgcaaataggactgcttcacagc
	275-277A-R	ctatttgcagctgcccttctcggcagtatacatag
	278-280A-F	ctggctgcagctgcttcacagctgagactg
	278-280A-R	agcagctgcagccagagccttcttctgcg
	282-283A-F	gcagcagctgagactgctgataaactac
	282-283A-R	gcagctcagctgctgcagcagctcattcagag
	285-286A-F	ggggctgctgataaactactttctggag
	285-286A-R	agtagtttatcagcagcccaagctgatgaagcagtc
	288-290A-F	gctgcagcactttctggagttcaggctatgtaag
	288-290A-R	aaactccagaaaagtgcctgcagcagcagctcagctgatgaag
	291-294A-F	gagactgctgataaactagctgctgcagtttcaggctatgtaagcaagcc
	291-294A-R	atagcctgaaactgcagcagctagtttatcagcagctcagctg
	295-298A-F	gagttgcagctgctgtaagcaagccaaactgtatg
	295-298A-R	cttacagcagctgcaactccagaaagtagtttatcagcag
	298-300A-F	ctatgcagctgcaccaaactgtatgaaatcgacg
	298-300A-R	gtttggctgcagctgcatagcctgaaactccagaaagtag
	301-303A-F	gcaaggcagctgcttatgaaatcgacgagataatggaagc
	301-303A-R	cataagcagctgcttgccttcatagcctgaaactcc
	304-306A-F	gtgctgcagctgacgagataatggaagctacgaag
	304-306A-R	cgtcagctgcagccacgtttggctgcttcatagc
	307-309A-F	gaaatcgcctgcagcaatggaagctacgaaggatttcag
	307-309A-R	ccattgctgcagcgatttcatacacgtttgctgc
	310-312A-F	cgagatagcagctgctacgaaggatttcagcg
	310-312A-R	gtagcagctgctatctcgtcgatttcatacacgtttg
	313-315A-F	gctgcagctgcattcagcgatgagtgcaagg
	313-315A-R	ctgaaigcagctgcagcttcattatctcgtcgatttc
	316-318A-F	ggatgcagctgctgagtgcaaggttggggaatc
	316-318A-R	ctcagcagctgcatccttcgtagcttcattatctc
	319-321A-F	cgatgctgcagctgttggggaatcagtgtaagc
	319-321A-R	caacagctgcagcatcgtgaaatccttcgtagc
	322-324A-F	gcaaggcagctgcatcagtgtaagccaacatag
	322-324A-R	gatgcagctgcttgcactcatcgtgaaatc
	325-327A-F	gaagcagctgcaaggccaacatagaaggtcg
	325-327A-R	gcctttgcagctgcttcccaaccttcgactc
	268-277DE-F	agaagcgcggatccgaattcaataggactgct
	268-277DE-R	ctgatgaagcagtcctattgaattcggatccgcgcttc
	278-287DE-F	gcagaaaagaaggctctggataaactactttctggagtttcagg
	278-287DE-R	aaactccagaaagtagtttatccagagccttcttctgcg
	298-307DE-F	caggctatgagataatggaagctacgaaggatttc
	298-307DE-R	cttcattatctcatagcctgaaactccagaaagtag

Supplemental Table 2. (continued)

	308-317DE-F	gaaatcgacgatgagtgcaagggtgggg
	308-317DE-R	ttgcactcatcgctcgatttcatacacgtttggc
	318-327DE-F	gatttcagcaaggccaacatagaaggtcg
	318-327DE-R	gttggccttgctgaaatccttcgtagcttcc
	328-337DE-F	tcagtgtacgtaagaaaaatcaaggagggtgggcc
	328-337DE-R	atfttctttacgtacactgattcccaaccttg
	288-294A-F	gcagctgccgcagcgggtgcctcaggctatgtaagcaagcc
	288-294A-R	gcagccgctgcggcagctgcagcagtcagctgatgaagc
	288-298PBS1-F	ggagacaaatctcatgtctcctcaggctatgtaagcaagcc
	288-298PBS1-R	ggagacatgagatttctcctcagcagtcagctgatgaagc
	N5-283S-R	tgaagcagtcctattcagagcc
	283Y-F	tctgaataggactgcttcatacgctgagactgc
	N5-294G	agaaagtagtttatcagcagtcctcag
	294Q-F	ctgctgataaactactttctcagggttcaggctatgtaag
	N5-303Y-R	cacgtttggcttgcttacatag
	304S-F	gcaagccaaacgtgtccgaaatcgacgag
	N5-310AT-R	ttccattatctcgctgatttcatacac
	311-2 IY-F	gaaatcgacgagataatggaaatctacaaggatttcagcgatgagtg

Supplemental Table 2. (continued)

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Reviewer #1 (Public Review):

Bacterial effectors that interfere with the inner molecular workings of eukaryotic host cells are of great biological significance across disciplines. On the one hand they help us to understand the molecular strategies that bacteria use to manipulate host cells. On the other hand they can be used as research tools to reveal molecular details of the intricate workings of the host machinery that is relevant for the interaction/defence/symbiosis with bacteria. The authors investigate the function and biological impact of a rhizobial effector that interacts with and modifies, and curiously is modified by, legume receptors essential for symbiosis. The molecular analysis revealed a bacterial effector that cleaves a plant symbiosis signaling receptor to inhibit signaling and the host counterplay by phosphorylation via a receptor kinase. These findings have potential implications beyond bacterial interactions with plants.

Bao and colleagues investigated how rhizobial effector proteins can regulate the legume root nodule symbiosis. A rhizobial effector is described to directly modify symbiosis-related signaling proteins, altering the outcome of the symbiosis. Overall, the paper presents findings that will have a wide appeal beyond its primary field.

Out of 15 identified effectors from *Sinorhizobium fredii*, they focus on the effector NopT, which exhibits proteolytic activity and may therefore cleave specific target proteins of the host plant. They focus on two Nod factor receptors of the legume *Lotus japonicus*, NFR1 and NFR5, both of which were previously found to be essential for the perception of rhizobial nod factor, and the induction of symbiotic responses such as bacterial infection thread formation in root hairs and root nodule development (Madsen et al., 2003, *Nature*; Tirichine et al., 2003; *Nature*). The authors present evidence for an interaction of NopT with NFR1 and NFR5. The paper aims to characterize the biochemical and functional consequences of these interactions and the phenotype that arises when the effector is mutated.

Evidence is presented that in vitro NopT can cleave NFR5 at its juxtamembrane region. NFR5 appears also to be cleaved in vivo. and NFR1 appears to inhibit the proteolytic activity of NopT by phosphorylating NopT. When NFR5 and NFR1 are ectopically over-expressed in leaves of the non-legume *Nicotiana benthamiana*, they induce cell death (Madsen et al., 2011, *Plant Journal*). Bao et al., found that this cell death response is inhibited by the coexpression

of nopT. Mutation of nopT alters the outcome of rhizobial infection in *L. japonicus*. These conclusions are well supported by the data.

The authors present evidence supporting the interaction of NopT with NFR1 and NFR5. In particular, there is solid support for cleavage of NFR5 by NopT (Figure 3) and the identification of NopT phosphorylation sites that inhibit its proteolytic activity (Figure 4C). Cleavage of NFR5 upon expression in *N. benthamiana* (Figure 3A) requires appropriate controls (inactive mutant versions) that have been provided, since *Agrobacterium* as a closely rhizobia-related bacterium, might increase defense related proteolytic activity in the plant host cells.

Key results from *N. benthamiana* appear consistent with data from recombinant protein expression in bacteria. For the analysis in the host legume *L. japonicus* transgenic hairy roots were included. To demonstrate that the cleavage of NFR5 occurs during the interaction in plant cells the authors build largely on western blots. Regardless of whether *Nicotiana* leaf cells or *Lotus* root cells are used as the test platform, the Western blots indicate that only a small proportion of NFR5 is cleaved when co-expressed with nopT, and most of the NFR5 persists in its full-length form (Figures 3A-D). It is not quite clear how the authors explain the loss of NFR5 function (loss of cell death, impact on symbiosis), as a vast excess of the tested target remains intact. It is also not clear why a large proportion of NFR5 is unaffected by the proteolytic activity of NopT. This is particularly interesting in *Nicotiana* in the absence of Nod factor that could trigger NFR1 kinase activity.

It is also difficult to evaluate how the ratios of cleaved and full-length protein change when different versions of NopT are present without a quantification of band strengths normalized to loading controls (Figure 3C, 3D, 3F). The same is true for the blots supporting NFR1 phosphorylation of NopT (Figure 4A).

It is clear that mutation of nopT results in a quantitative infection phenotype. Nodule primordia and infection threads are still formed when *L. japonicus* plants are inoculated with Δ nopT mutant bacteria, but it is not clear if these primordia are infected or develop into fully functional nodules (Figure 5). A quantification of the ratio of infected and non-infected nodules and primordia would reveal whether NopT is only active at the transition from infection focus to thread or perhaps also later in the bacterial infection process of the developing root nodule.

<https://doi.org/10.7554/eLife.97196.1.sa1>

Reviewer #2 (Public Review):

Summary:

This manuscript presents data demonstrating NopT's interaction with Nod Factor Receptors NFR1 and NFR5 and its impact on cell death inhibition and rhizobial infection. The identification of a truncated NopT variant in certain *Sinorhizobium* species adds an interesting dimension to the study. These data try to bridge the gaps between classical Nod-factor-dependent nodulation and T3SS NopT effector-dependent nodulation in legume-rhizobium symbiosis. Overall, the research provides interesting insights into the molecular mechanisms underlying symbiotic interactions between rhizobia and legumes.

Strengths:

The manuscript nicely demonstrates NopT's proteolytic cleavage of NFR5, regulated by NFR1 phosphorylation, promoting rhizobial infection in *L. japonicus*. Intriguingly, authors also identify a truncated NopT variant in certain *Sinorhizobium* species, maintaining NFR5 cleavage but lacking NFR1 interaction. These findings bridge the T3SS effector with the

classical Nod-factor-dependent nodulation pathway, offering novel insights into symbiotic interactions.

Weaknesses:

(1) In the previous study, when transiently expressed NopT alone in *Nicotiana tobacco* plants, proteolytically active NopT elicited a rapid hypersensitive reaction. However, this phenotype was not observed when expressing the same NopT in *Nicotiana benthamiana* (Figure 1A). Conversely, cell death and a hypersensitive reaction were observed in Figure S8. This raises questions about the suitability of the exogenous expression system for studying NopT proteolysis specificity.

(2) NFR5 Loss-of-function mutants do not produce nodules in the presence of rhizobia in lotus roots, and overexpression of NFR1 and NFR5 produces spontaneous nodules. In this regard, if the direct proteolysis target of NopT is NFR5, one could expect the NGR234's infection will not be very successful because of the Native NopT's specific proteolysis function of NFR5 and NFR1. Conversely, in Figure 5, authors observed the different results.

(3) In Figure 6E, the model illustrates how NopT digests NFR5 to regulate rhizobia infection. However, it raises the question of whether it is reasonable for NGR234 to produce an effector that restricts its own colonization in host plants.

(4) The failure to generate stable transgenic plants expressing NopT in *Lotus japonicus* is surprising, considering the manuscript's claim that NopT specifically proteolyzes NFR5, a major player in the response to nodule symbiosis, without being essential for plant development.

<https://doi.org/10.7554/eLife.97196.1.sa0>

Author response:

eLife assessment

This valuable study reveals how a rhizobial effector protein cleaves and inhibits a key plant receptor for symbiosis signaling, while the host plant counters by phosphorylating the effector. The molecular evidence for the protein-protein interaction and modification is solid, though biological evidence directly linking effector cleavage to rhizobial infection is incomplete. With additional functional data, this work could have implications for understanding intricate plant-microbe dynamics during mutualistic interactions.

Thank you for this helpful comment. In the revised manuscript version, we will be more prudent with directly linking cleavage of Nod factor receptors by NopT and rhizobial infection.

We plan to modify the Title, the One-Sentence Summary, Abstract, and Discussion regarding this point.

Public Reviews:

Reviewer #1 (Public Review):

Bacterial effectors that interfere with the inner molecular workings of eukaryotic host cells are of great biological significance across disciplines. On the one hand they help us to understand the molecular strategies that bacteria use to manipulate host cells. On the other hand they can be used as research tools to reveal molecular details of the intricate workings of the host machinery that is relevant for the interaction/defence/symbiosis

with bacteria. The authors investigate the function and biological impact of a rhizobial effector that interacts with and modifies, and curiously is modified by, legume receptors essential for symbiosis. The molecular analysis revealed a bacterial effector that cleaves a plant symbiosis signaling receptor to inhibit signaling and the host counterplay by phosphorylation via a receptor kinase. These findings have potential implications beyond bacterial interactions with plants.

Thank you for highlighting the broad significance of rhizobial effectors in understanding legume-rhizobium interactions. We fully agree with your assessment and will emphasize these points in the revised Introduction and Discussion sections of our manuscript. Specifically, we will expand our Discussion regarding the potential impact of the NopT interaction with symbiotic receptor kinases on plant immune signaling and regarding the general significance of our work.

Bao and colleagues investigated how rhizobial effector proteins can regulate the legume root nodule symbiosis. A rhizobial effector is described to directly modify symbiosis-related signaling proteins, altering the outcome of the symbiosis. Overall, the paper presents findings that will have a wide appeal beyond its primary field.

*Out of 15 identified effectors from *Sinorhizobium fredii*, they focus on the effector NopT, which exhibits proteolytic activity and may therefore cleave specific target proteins of the host plant. They focus on two Nod factor receptors of the legume *Lotus japonicus*, NFR1 and NFR5, both of which were previously found to be essential for the perception of rhizobial nod factor, and the induction of symbiotic responses such as bacterial infection thread formation in root hairs and root nodule development (Madsen et al., 2003, *Nature*; Tirichine et al., 2003; *Nature*). The authors present evidence for an interaction of NopT with NFR1 and NFR5. The paper aims to characterize the biochemical and functional consequences of these interactions and the phenotype that arises when the effector is mutated.*

Thank you for your positive feedback on our manuscript. In the revised Introduction and Discussion sections, we plan to better emphasize the interdisciplinary significance of our work. We will show how the knowledge gained from our study can contribute to a better understanding of microbial interactions with eukaryotic hosts in general, which may have a stimulating effect on future research in various research areas such as pathogenesis and immunity.

To ensure that the readers can easily follow the rationale behind our experiments, we will improve the Results section and provide more detailed explanations of how NopT among 15 examined effectors was selected. Additionally, we will provide more background information on NopT and the roles of NFR1 and NFR5 in symbiotic signaling in the Introduction section. As suggested, we will include the references Madsen et al. (2003) and Tirichine et al. (2003) as well as additional references on rhizobial NopT proteins into our revised manuscript version.

*Evidence is presented that in vitro NopT can cleave NFR5 at its juxtamembrane region. NFR5 appears also to be cleaved in vivo. and NFR1 appears to inhibit the proteolytic activity of NopT by phosphorylating NopT. When NFR5 and NFR1 are ectopically over-expressed in leaves of the non-legume *Nicotiana benthamiana*, they induce cell death (Madsen et al., 2011, *Plant Journal*). Bao et al., found that this cell death response is inhibited by the coexpression of *nopT*. Mutation of *nopT* alters the outcome of rhizobial infection in *L. japonicus*. These conclusions are well supported by the data.*

We appreciate that you recognize the value of our data.

*The authors present evidence supporting the interaction of NopT with NFR1 and NFR5. In particular, there is solid support for cleavage of NFR5 by NopT (Figure 3) and the identification of NopT phosphorylation sites that inhibit its proteolytic activity (Figure 4C). Cleavage of NFR5 upon expression in *N. benthamiana* (Figure 3A) requires appropriate controls (inactive mutant versions) that have been provided, since *Agrobacterium* as a closely rhizobia-related bacterium, might increase defense related proteolytic activity in the plant host cells.*

Thank you for recognizing the use of an inactive NopT variant in Figure 3A. In fact, increased activity of plant proteases induced by *Agrobacterium* is an important point that should not be neglected. We plan to mention this aspect in our revised Discussion.

In the context of your comments, we are planning to make the following improvements to the manuscript:

- (1) We will add a more detailed description of the experimental conditions under which the cleavage of NFR5 by NopT was observed in vitro and in vivo.
- (2) We plan to provide more comprehensive data on the phosphorylation of NopT by NFR1, including phosphorylation assays and mass spectrometry results. These additional data support the proposed mechanism by which NFR1 inhibits the proteolytic activity of NopT.
- (3) We will expand the Discussion on the cell death response induced by ectopic expression of NFR1 and NFR5 in *Nicotiana benthamiana*. We will include more details from Madsen et al. (2011) to contextualize our findings with published literature.

We believe these additions and clarifications will enhance the clarity and impact of our findings.

*Key results from *N. benthamiana* appear consistent with data from recombinant protein expression in bacteria. For the analysis in the host legume *L. japonicus* transgenic hairy roots were included. To demonstrate that the cleavage of NFR5 occurs during the interaction in plant cells the authors build largely on western blots. Regardless of whether *Nicotiana* leaf cells or *Lotus* root cells are used as the test platform, the Western blots indicate that only a small proportion of NFR5 is cleaved when co-expressed with nopT, and most of the NFR5 persists in its full-length form (Figures 3A-D). It is not quite clear how the authors explain the loss of NFR5 function (loss of cell death, impact on symbiosis), as a vast excess of the tested target remains intact. It is also not clear why a large proportion of NFR5 is unaffected by the proteolytic activity of NopT. This is particularly interesting in *Nicotiana* in the absence of Nod factor that could trigger NFR1 kinase activity.*

Thank you for your comments regarding the cleavage of NFR5 and its functional implications. In the revised version, we will change our manuscript taking into account the following considerations:

- (1) We acknowledge that the Western blots indicate only a small proportion of NFR5 is cleaved when co-expressed with NopT. It is worth noting in this context that the proteins were expressed at high levels which likely do not reflect the natural situation in *L. japonicus*. Low production of cleaved NFR5 in our Western blots with transformed *N. benthamiana* or *L. japonicus* cells thus may simply reflect an experimental effect due to high NFR5 protein synthesis. We suggest that the presence of high amounts of intact NFR5 does not have a significant functional impact on plant responses (cell death in *N. benthamiana*, rhizobial infection of *L. japonicus*) whereas NFR5 cleavage (or formation of NFR5 cleavage products) may be crucial for the observation of the observed phenotypic changes. The fraction of

cleaved NFR5, although small, may be sufficient to disrupt crucial signaling pathways, leading to observable phenotypic changes. We will address possible differences between experimental and natural protein levels in our revised Discussion.

(2) We studied in our work three biochemical aspects of NopT: (i) physical binding of NopT to NFR1 and NFR5 (ii) proteolytical cleavage of NFR5 by NopT and (iii) phosphorylation of NopT by NFR1. These three biochemical properties appear to influence each other. Phosphorylation of NopT by NFR1 appears to reduce its proteolytic activity, thereby counteracting NFR5 degradation by NopT (NFR5 homeostasis). Moreover, as NopT is a phosphorylation substrate for NFR1, NopT probably interferes with kinase mediated downstream responses of NFR1. Thus, NFR5 cleavage activity of NopT appears to be only one feature of NopT. We plan to mention these considerations in our revised Discussion.

It is also difficult to evaluate how the ratios of cleaved and full-length protein change when different versions of NopT are present without a quantification of band strengths normalized to loading controls (Figure 3C, 3D, 3F). The same is true for the blots supporting NFR1 phosphorylation of NopT (Figure 4A).

Thank you for pointing out this aspect. Following your recommendation, we will quantify the band intensities for cleaved and full-length NFR5 in the experiments with different versions of NopT. These values will be normalized to loading controls. Similarly, the Western blots supporting NFR1 phosphorylation of NopT will be quantified. The data for normalized band intensities will be included into the revised figures. The quantifications will provide a clearer understanding of how the ratios of cleaved to full-length proteins change with different NopT variants and also will provide information to which extent NopT is phosphorylated by NFR1.

*It is clear that mutation of nopT results in a quantitative infection phenotype. Nodule primordia and infection threads are still formed when *L. japonicus* plants are inoculated with Δ nopT mutant bacteria, but it is not clear if these primordia are infected or develop into fully functional nodules (Figure 5). A quantification of the ratio of infected and non-infected nodules and primordia would reveal whether NopT is only active at the transition from infection focus to thread or perhaps also later in the bacterial infection process of the developing root nodule.*

Thank you for pointing this out. In the revised version of our manuscript, we will provide data showing that there are no obvious differences in nodule formation in plants inoculated with Δ nopT and wild-type NGR234, respectively. However, quantification of infection threads containing our GFP-labeled rhizobia in primordia and nodules would be difficult to perform due to strong autofluorescence signals in these tissues. The main goal of our study was to identify and characterize the interaction between NopT and Nod factor receptors. We therefore believe that an in-depth analysis of the bacterial infection process at later symbiotic stages is out of the scope of the present work.

Reviewer #2 (Public Review):

Summary:

This manuscript presents data demonstrating NopT's interaction with Nod Factor Receptors NFR1 and NFR5 and its impact on cell death inhibition and rhizobial infection. The identification of a truncated NopT variant in certain Sinorhizobium species adds an interesting dimension to the study. These data try to bridge the gaps between classical Nod-factor-dependent nodulation and T3SS NopT effector-dependent nodulation in legume-rhizobium symbiosis. Overall, the research provides interesting insights into the molecular mechanisms underlying symbiotic interactions between rhizobia and legumes.

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*The manuscript nicely demonstrates NopT's proteolytic cleavage of NFR5, regulated by NFR1 phosphorylation, promoting rhizobial infection in *L. japonicus*. Intriguingly, authors also identify a truncated NopT variant in certain *Sinorhizobium* species, maintaining NFR5 cleavage but lacking NFR1 interaction. These findings bridge the T3SS effector with the classical Nod-factor-dependent nodulation pathway, offering novel insights into symbiotic interactions.*

We appreciate that you recognize the value of our manuscript.

Weaknesses:

*(1) In the previous study, when transiently expressed NopT alone in *Nicotiana tabacum* plants, proteolytically active NopT elicited a rapid hypersensitive reaction. However, this phenotype was not observed when expressing the same NopT in *Nicotiana benthamiana* (Figure 1A). Conversely, cell death and a hypersensitive reaction were observed in Figure S8. This raises questions about the suitability of the exogenous expression system for studying NopT proteolysis specificity.*

We appreciate your attention to these plant-specific differences. In view of your comments, we plan to revise the Discussion and explain the different expression systems used for studying NopT effects *in planta*. Previous studies showed that NopT expressed in tobacco (*N. tabacum*) or in specific *Arabidopsis thaliana* ecotypes (with *PBS1/RPS5* genes) causes rapid cell death (Dai et al. 2008; Khan et al. 2022). Our data shown in Fig. S8 confirm these findings. As cell death (effector triggered immunity) is usually associated with induction of protease activities, we considered *N. tabacum* and *A. thaliana* plants as not suitable for testing NFR5 cleavage by NopT. In fact, no NopT/NFR5 experiments were performed with these plants in our study. In contrast, the expression of NopT in *Nicotiana benthamiana* did not lead to cell death in our experiments. Khan et al. 2022 also reported that cell death does not occur in *N. benthamiana* unless the cells were transformed with *PBS1/RPS5* constructs. Thus, *N. benthamiana* is a suitable expression system to analyze NopT protease activity on co-expressed substrates. Our revision aims to better understand the advantages of the *N. benthamiana* expression system for studying NopT mediated proteolysis of NFR5.

(2) NFR5 Loss-of-function mutants do not produce nodules in the presence of rhizobia in lotus roots, and overexpression of NFR1 and NFR5 produces spontaneous nodules. In this regard, if the direct proteolysis target of NopT is NFR5, one could expect the NGR234's infection will not be very successful because of the Native NopT's specific proteolysis function of NFR5 and NFR1. Conversely, in Figure 5, authors observed the different results.

Our inoculation experiments clearly show that NopT of NGR234 has a negative effect on formation of infection foci (Fig. 5A) and nodule primordia (Fig. 5E). Our biochemical analysis indicates that NopT targets the NFR1/NFR5 complex, which most likely impairs activation of downstream responses such as *NIN* gene expression. Accordingly, *NIN* promoter activity was found to be higher in roots inoculated with the $\Delta nopT$ mutant as compared to the NGR234 wild-type (Fig. 5B and 5D). It is therefore plausible that NopT impairs rhizobial infection of *L. japonicus* due to inhibition of NFR1/NFR5 functions. We agree with this Reviewer that it can be expected that “NGR234's infection will not be very successful”. Fig. 5 confirms that $\Delta nopT$ mutant is indeed a better symbiont and we do not think that we obtained “unexpectedly different results”. In the revised version, we will try to formulate our discussion text better in order to avoid any misunderstandings. Furthermore, will write as figure title “NopT dampens

rhizobial infection...” instead of “NopT regulates rhizobial infection...”. We are also considering changing the title of our manuscript.

(3) *In Figure 6E, the model illustrates how NopT digests NFR5 to regulate rhizobia infection. However, it raises the question of whether it is reasonable for NGR234 to produce an effector that restricts its own colonization in host plants.*

We acknowledge the potential paradox of NGR234 producing an effector that appears to restrict its own colonization in host plants. In fact, depending on the host plant, most rhizobial effectors are “double-edged swords” that play either a positive or negative role in the symbiosis. In response to your comment, we will discuss the possibility that NopT may confer selective advantages in interactions between NGR234 and host plants where NopT plays a positive symbiotic role (Dai et al. 2008; Kambara et al. 2009). Inhibition of NFR1/NFR5 functions by NopT in these host plants could be a feedback response in cells in which symbiotic signaling has already started. It is tempting to speculate that the interaction between NopT and Nod factor receptors reduces Nod factor perception and downstream signaling to avoid a possible overreaction of symbiotic signaling, which may result in hypernodulation or formation of empty nodules without bacteria. Furthermore, it is tempting to speculate that NopT targets not only Nod factor receptors but also other host proteins to promote symbiosis, e.g. by suppressing excessive immune responses triggered by hyperinfection of rhizobia. In our revised manuscript, we will highlight the need for further investigations to elucidate the precise mechanisms underlying the observed infection phenotype and the role of NopT in modulating symbiotic signaling pathways.

(4) *The failure to generate stable transgenic plants expressing NopT in Lotus japonicus is surprising, considering the manuscript's claim that NopT specifically proteolyzes NFR5, a major player in the response to nodule symbiosis, without being essential for plant development.*

Thank you for your comments. The failure to obtain *L. japonicus* plants constitutively expressing NopT was indeed surprising and suggests that NopT targets not only NFR5 but also other proteins in *L. japonicus*. The number of NopT substrates in plants could be greater than assumed. For example, we show in our work that NopT can cleave AtLYK5 and LjLYS11. In our manuscript, we don't provide protocols and data on our efforts to construct *L. japonicus* plants stably expressing NopT. Indeed, it cannot be completely ruled out that the observed failure is not due to NopT expression, but rather to other factors that influence the transformation and regeneration of explants into whole plants. Our results should therefore not be over-interpreted. We consider a discussion of our failed transformation experiments to be somewhat preliminary and not central to this manuscript. Therefore, we plan to modify our Discussion and delete the sentence reporting that stable transgenic plants expressing NopT have not been successfully generated.

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